ACADEMIC REGULATIONS

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SYLLABUS

Effective from the Academic Year: 2016-17

Faculty of Pharmacy

Bachelor of Pharmacy Programme

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Pharmacy

ACADEMIC REGULATIONS

Bachelor of Pharmacy (B. Pharm.) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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CHARUSAT

FACULTY OF PHARMACY
ACADEMIC REGULATIONS
Bachelor of Pharmacy (B. Pharm.) Programme

To ensure uniform system of education, duration of undergraduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between courses and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across The Charotar University of Science and Technology (CHARUSAT) at Undergraduate levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate programme	(B.Pharm)
Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions

As decided from time to time by the Government of Gujarat and CHARUSAT

4. Mode of admissions

As decided from time to time by the Government of Gujarat and CHARUSAT

5. Programme structure and Credits

Please refer detailed syllabus

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the B. Pharm. Programme. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member (s) based on continuous assessment, for 20%* of the marks for the course; and

7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 80%* of the marks for the course.

**as per norms laid down by Pharmacy Council of India (PCI)*

7.1.3 **Theory and Practical component of the same course shall be considered as separate courses.**

7.2 Internal Evaluation (Theory)

Please review Page No. 20.

7.3 Internal evaluation (Practicals)

The distribution of marks for calculating the internal marks in every course of practical shall be:

Performance of the exercise	20 Marks (One test of three hours duration)
Viva	20 Marks (Two tests each of 10marks)
Quiz	20 Marks (Two tests each of 10marks)
Record (Journal)	20 Marks
Total	80 Marks
Internal Marks (out of 20)	(Marks obtained out of 80) / 4

7.4 University Examination

7.4.1. The final examination by the University for 80% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.4.2. In order to earn the credit in a course a student has to obtain grade other than FF.

7.5 Performance at Examination

7.5.1 Minimum performance with respect to university examination as well as overall (university + internal) will be an important consideration for passing a course.

Details of minimum percentage of marks to be obtained in the examinations is as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

7.5.2 If a candidate obtains minimum required marks per course in university examination but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.(As per the clause 7.5.1)

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade **as well as to** a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Numerical point (Grade Point) corresponding to the letter grade	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

(iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

(v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Award of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He/She should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He/She should have cleared external and overall evaluation components in every course; and

9.1.3 He/She should have secured a minimum CGPA of 5.0 at the end of the programme;

9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.1.5 The student is required to undergo practical training for a period not less than 150 hours / 5 weeks, during vacation due at the end of 4^{th} Semester B.Pharm and /or 6^{th} Semester B.Pharm and/or at the end of 8^{th} Semester B.Pharm in a Pharmaceutical Industry/Tertiary Care Hospital approved the Institute

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10 Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme

Distinction:	CGPA ≥ 7.5
First class:	CGPA ≥ 6.00 & < 7.49
Second Class:	CGPA ≥ 5.0 & < 5.99
Pass Class:	< CGPA 5.00

11 Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA,CGPA, class obtained, etc.

Vision of RPCP

To become a premier pharma institute by creating world class pharmacists and researchers for dissemination and creation of knowledge

Mission of RPCP

To develop entrepreneurial pharmacist with a penchant for research and development

Programme Educational Objective

1. To produce competent pharmacy graduates who shall be able to secure their career in pharmaceutical industry and/or institution of higher studies by dissemination of knowledge and training in pharmacy.
2. To acquire the knowledge about advances in pharmacy and to orient towards socially relevant research and development.
3. To encourage students for self-learning through modern information resources.
4. To enhance communication and critical thinking abilities for effective delivery of societal responsibilities.
5. To nurture and facilitate entrepreneurial capabilities and opportunities.

PROGRAM OUTCOMES

1. Pharmacy Knowledge: Possess knowledge and comprehension of the core and basic knowledge associated with the profession of pharmacy, including biomedical sciences; pharmaceutical sciences; behavioral, social, and administrative pharmacy sciences; and manufacturing practices.

2. Planning Abilities: Demonstrate effective planning abilities including time management, resource management, delegation skills and organizational skills. Develop and implement plans and organize work to meet deadlines.

3. Problem analysis: Utilize the principles of scientific enquiry, thinking analytically, clearly and critically, while solving problems and making decisions during daily practice. Find, analyze, evaluate and apply information systematically and shall make defensible decisions.

4. Modern tool usage: Learn, select, and apply appropriate methods and procedures, resources, and modern pharmacy-related computing tools with an understanding of the limitations.

5. Leadership skills: Understand and consider the human reaction to change, motivation issues, leadership and team-building when planning changes required for fulfillment of practice, professional and societal responsibilities. Assume participatory roles as responsible citizens or leadership roles when appropriate to facilitate improvement in health and wellbeing.

6. Professional Identity: Understand, analyze and communicate the value of their professional roles in society (e.g. health care professionals, promoters of health, educators, managers, employers, employees).

7. Pharmaceutical Ethics: Honour personal values and apply ethical principles in professional and social contexts. Demonstrate behavior that recognizes cultural and personal variability in values, communication and lifestyles. Use ethical frameworks; apply ethical principles while making decisions and take responsibility for the outcomes associated with the decisions.

8. Communication: Communicate effectively with u the pharmacy community and with society at large, such as, being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.

9. The Pharmacist and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and the consequent responsibilities relevant to the professional pharmacy practice.

10. Environment and sustainability: Understand the impact of the professional pharmacy solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

11. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. Self-assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

Semester-7
Scheme of Teaching

Course Code	Course Name	Contact hours per week		Credit	
		Theory	Practical	Theory	Practical
PH430	Clinical Pharmacy and Pharmacotherapeutics-I	3	3	3	1
PH431	Traditional Medicine and Herbal Drugs Technology	3	3	3	2
PH432	Pharmaceutical Analysis-III	3	3	3	2
PH418	Medicinal Chemistry-III	3	---	3	---
PH433	Pharmaceutical Technology-II	3	3	3	2
PH420.1	Pharmacy Practice - II	2	---	2	---
PH420.2	Herbal Formulations- II	2	---	2	---
PH420.3	Production Of Bulk Drugs And Drug Intermediates II	2	---	2	---
PH420.4	Pharmaceutical Quality Assurance And Quality Control - II	2	---	2	---
PH420.5	Drug Regulatory Affairs- II	2	---	2	---
PH420.6	Sterile Product Technology- II	2	---	2	---
PH421	Environmental Sciences	2	---	2	---
	Total	19	12	19	7
Total credits: 26					

Scheme of Evaluation

Course Code	Course Name	Theory			Practical		
		University	Institute	Total	University	Institute	Total
PH430	Clinical Pharmacy and Pharmacotherapeutics-I	80	20	100	80	20	100
PH431	Traditional Medicine and Herbal Drugs Technology	80	20	100	80	20	100
PH432	Pharmaceutical Analysis-III	80	20	100	80	20	100
PH418	Medicinal Chemistry-III	80	20	100	---	---	---
PH433	Pharmaceutical Technology-II	80	20	100	80	20	100
PH420.1	Pharmacy Practice - II	50	50	100	---	---	---
PH420.2	Herbal Formulations- II	50	50	100	---	---	---
PH420.3	Production Of Bulk Drugs And Drug Intermediates II	50	50	100	---	---	---
PH420.4	Pharmaceutical Quality Assurance And Quality Control - II	50	50	100	---	---	---
PH420.5	Drug Regulatory Affairs- II	50	50	100	---	---	---
PH420.6	Sterile Product Technology- II	50	50	100	---	---	---
PH421	Environmental Sciences	---	50	50	---	---	---
	Total	450	200	650	320	80	400
Total marks: 1050							

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
BACHELOR OF PHARMACY (B. Pharm.) PROGRAMME
From Academic year: 2016-17

Semester-8
Scheme of Teaching

Course Code	Course Name	Contact hours per week			Credit		
		Theory	Practical	Project	Theory	Practical	Project
PH434	Clinical Pharmacy and Pharmacotherapeutics-II	3	3	--	3	2	--
PH435	Medicinal Chemistry-IV	3	3	--	3	1	--
PH436	Dosage Form Design	3	3	--	3	2	--
PH425	Novel Drug Delivery Systems	2	--	--	2	--	--
PH437.1	Pharmacy Practice - III	--	3	3	--	3	1
PH437.2	Herbal Formulations- III	--	3	3	--	3	1
PH437.3	Production Of Bulk Drugs And Drug Intermediates III	--	3	3	--	3	1
PH437.4	Pharmaceutical Quality Assurance And Quality Control - III	--	3	3	--	3	1
PH437.5	Drug Regulatory Affairs-III	--	3	3	--	3	1
PH437.6	Sterile Product Technology - III	--	3	3	--	3	1
PH438	Entrepreneurship for pharma professionals	3	3	--	3	1	--
	Total	**14**	**15**	**3**	**14**	**9**	**1**
Total credits: 24							

Scheme of evalaution

Course Code	Course Name	Theory			Practical			Project		
		University	Institute	Total	University	Institute	Total	University	Institute	Total
PH434	Clinical Pharmacy and Pharmacotherapeutics-II	80	20	100	80	20	100	--	--	--
PH435	Medicinal Chemistry-IV	80	20	100	80	20	100	--	--	--
PH436	Dosage Form Design	80	20	100	80	20	100	--	--	--
PH425	Novel Drug Delivery Systems	80	20	100	---	---	---	--	--	--
PH437.1	Pharmacy Practice - III	---	---	---	50	50	100	50	50	100
PH437.2	Herbal Formulations-III	---	---	---	50	50	100	50	50	100
PH437.3	Production Of Bulk Drugs And Drug Intermediates III	---	---	---	50	50	100	50	50	100
PH437.4	Pharmaceutical Quality Assurance And Quality Control - III	---	---	---	50	50	100	50	50	100
PH437.5	Drug Regulatory Affairs- III	---	---	---	50	50	100	50	50	100
PH437.6	Sterile Product Techonology - III	---	---	---	50	50	100	50	50	100
PH438	Entrepreneurship for pharma professionals	80	20	100	80	20	100	---	---	---
	Total	400	100	500	370	130	500	50	50	100
Total marks: 1100										

Instructional Methods and Pedagogy:

The course will emphasize active classroom interaction based on students' prior preparation. It will also involve little self-learning components. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students come prepared for the classes. Faculty member/s shall explain in the classroom using traditional methods, multimedia projector and via video clippings.

Evaluation:

Internal Evaluation –– Theory Courses

1. **The performance of each student in each theory course will be evaluated as follows:**
   1.1. Internal evaluation by the course faculty member (s) based on continuous assessment, for 20%* of the marks for the course; and
   1.2. Final examination by the University through written paper of every theory course as per the scheme prescribed by the University, for 80%* of the marks for the course. [*as per norms laid down by Pharmacy Council of India (PCI)]

2. Internal evaluation for Theory courses shall be as per following scheme:

Sr. No.	Component	Marks	Duration
1	Unit test – 1	20	30 minutes
2	Unit test – 2	20	30 minutes
3	Assignment	30	45 minutes for reading + 75 minutes for answering
4	Theory test	30	75 minutes
	Total marks	100	--
	Internal marks (Out of 20)	Marks obtained out of 100/5	--

The internal evaluation shall be conducted as per the schedule notified by the Institute for every course in a semester.

1. **Unit tests**
   1.1. The questions for unit tests will be of objective types. It may be of multiple choice questions (MCQs), fill in the blanks, match the followings or any other type of objective question.
   1.2. The unit tests may be paper based or computer based.
   1.3. All questions are compulsory, no options to be given. Each question is of one mark.
2. **Assignments**
   2.1. Questions shall be descriptive type.
   2.2. Reading material of 10-15 A4 size pages will be given to the students at the starting of evaluation period.

3. **Theory test**
   3.1. The questions for Theory test will be of descriptive type (Long answer questions, short notes etc.).

Internal Evaluation – Bachelor of Pharmacy (B. Pharm.) Program –Practical Courses

1. **The performance of each student in each practical course will be evaluated as follows:**
   1.1. Internal evaluation by the course faculty member (s) based on continuous assessment, for 20%* of the marks for the course; and
   1.2. Final examination by the University through practical test or oral test of every practical course as per the scheme prescribed by the University, for 80%* of the marks for the course. [**as per norms laid down by Pharmacy Council of India (PCI)*]

2. Internal evaluation for Theory courses shall be as per following scheme:

Sr. No.	Component	Marks	Remarks
1	Performance of the exercise	20	One test of three hours duration
2	Viva	20	Two tests each of 10marks
3	Quiz	20	Two tests each of 10marks
4	Record (Journal)	20	--
	Total marks	80	--
	Internal marks (Out of 20)	(Marks obtained out of 80)/4	--

Scheme of Evaluation for Electives:

The performance of student in each elective course shall be evaluated based on continuous assessment (Internal Evaluation, 50 %) at Institute level evaluation and end semester examination (University Evaluation, 50%) by the university

Sem.	Theory			Practical			Project		
	Internal	University	Total	Internal	University	Total	Internal	University	Total
6th	50	50	100	--	---	---	--	---	---
7th	50	50	100	--	---	---	--	---	---
8th	---	---	--	50	50	100	50	50	100

Internal Evaluation –Theory

1. The internal evaluation of theory courses shall be conducted as per following scheme:

Sr.No	Test	Duration (Min)	Marks
1	Unit Test - I	30	15
2	Unit Test - 2	30	15
3	Assignment	50	20
		Total	50

1) Unit Test – I
   i) It shall be of multiple choice questions (MCQs) only
   ii) It may be paper based or computer based
   iii) All questions are compulsory, no options to be given. Each question shall be of one mark.
2) Unit Test – II
   i) It shall be of objective type of questions
   ii) It may be paper based or computer based
   iii) All questions are compulsory, no options to be given. Each question shall be of one mark.
3) Assignment
   i) It shall be based on cases study analysis, problem based questions or seminar evaluation
   ii) In case of seminar based assignment the evaluation shall be carried out, based on power point presentation – 10 marks and viva – voce -10 marks
   iii) In case of case studies and problem based questions, written theory test of total 20 marks shall be conducted.

Internal Evaluation Practical

The internal evaluation of practical course shall be conducted as pr following scheme:

Sr.No	Test	Duration (Min)	Marks
1	Quiz - I	10	05
2	Quiz - 2	10	05
3	Viva - I	10	05
4	Viva – 2	10	05
5	Performance of Practical/Case Analysis	90	25
6	Laboratory Record (Journal)	---	05
		Total	50

Project Evaluation- Internal and University Evaluation

1. Project shall be evaluated based on power point presentation and viva-voce at Institute level and University level examination
2. Students shall submit hard copy of the project at the time of University Examination.

SYLLABI

(Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CLINICAL PHARMACY & PHARMACOTHERAPEUTICS-I (PH430)

(Theory & Practical)

Credits: 3 (Theory) | **Contact Hours per week: 3 (Theory)**
: 1 (Practical) | **3(Practical)**

Course Outcome:

The student will be able:

- To describe the concept of Clinical Pharmacy & Pharmacotherapy.
- To describe role of clinical pharmacist as a part of health care team.
- To describe the theoretical considerations and basic principles of pharmacotherapeutics in the management of the diseases and conditions of various body system such as CVS, renal , hematology and malignant system with the help of standard treatment guidelines and treatment algorithms..
- To recognize the diagnostic criteria required to specific disease as well as any clinical testing requirements for monitoring drug effectiveness and potential toxicity.

Outline of the Course (Theory):

Sr. No.	Unit	Minimum No. of Contact Hours	Approx. Weightage %
1	Introduction to Clinical Pharmacy	2	4
2	Basic concept of Pharmacotherapy	3	5
3	Drug interactions	2	6
4	General Principles of Clinical toxicology	2	4
5	Therapeutic Drug Monitoring	3	10
6	a) Pharmacovigilance b) Pharmacoepidemiology	4	10
7	Clinical laboratory tests and their interpretation	4	10
8	Clinical Presentation, Diagnosis, Complication & Management of following Diseases/Disorders: a) Cardiovascular Disorders	13	27
	b) Renal Disorders	5	10
	c) Hematological Disorders	3	6
	d) Malignant Disorders	4	8
Total		45	100

Detailed syllabus (theory):

Sr. No.	Name of Chapter	References	Hrs. required
1	Introduction to Clinical Pharmacy: Development and scope of clinical pharmacy, concept of health care team, Role and Functions of clinical pharmacist as a member of health care team.	1,4,6	2
2	Basic concept of Pharmacotherapy : Recording of medication history, self medication, nonprescription drug usage, Communication skills of Clinical Pharmacist - Behavioral and interpersonal, with patients and other professionals	1,4	3
3	Drug interactions: Prescription monitoring, documentation and methods for minimizing clinically relevant drug interactions.	15,3,9	2
4	General Principles of Clinical toxicology	1,3,9,20	2
5	Therapeutic Drug Monitoring for safety and efficacy (medication chart review, clinical review, pharmacist interventions)	4,6,20	3
6	a) Pharmacovigilance. - The role of clinical pharmacist in monitoring and prevention of Adverse Drug Reactions b)Concepts of Pharmacoepidemiology	12,13,17,18	4
7	Interpretation of clinical laboratory tests: Hematological, pathological and biochemical investigations as markers of Major organ damage and their effect on drug therapy decisions	5,7,8,9	4
8	Clinical presentation, Diagnosis, Complication & Management of following Diseases/Disorders. a) Cardiovascular Disorders: Hypertension, Angina Pectoris Myocardial Infarction, Congestive Heart failure, Cardiac arrhythmias, Atherosclerosis, Stroke	1,8,9,10, 16, 20	13
	b) Renal Disorders: acute renal failure, chronic renal failure, acid-base disorders, Drug therapy individualization of patient with renal failure	1,5,8,9,10, 19,20	5
	c) Hematological Disorders: Anemia.	1,5,8,9,10,20	3
	d) Malignant Disorders: ,Leukemia, Lymphoma and mammary carcinomas	1,5,8,9,10,20	4

Detailed syllabus (Practical):

For clinical pharmacy related practical, Hypothetical cases and other necessary information like Product package insert, literatures etc will be given.

1	Basic introduction to Clinical Pharmacy & its role Introduction to Drug information Center
2	Study of Literature audit
3	Pharmacoepidemiology and Pharmacovigilance related case studies.
4	Clinical laboratory tests & their interpretations (Actual/ Hypothetical cases)
5	Case studies on Drug interactions
6	Case studies on Therapeutic Drug Monitoring
7	Case studies based on Cardiovascular, Renal, Hematological and Malignant Disorders.

Recommended study materials: (Latest edition)

1. Elements of Clinical Pharmacy: R.K Goyal, Parloop A.Bhatt & Mahesh D.Burande, B.S. Shah Prakashan.
2. Davidson's Principles & Practice of Medicines:Churchill Livingstone Elsevier.
3. Handbook of Drug Interactions, A clinical & forensic Guide by Ashraf Mozayani: Humana Press.
4. A Textbook of Clinical Pharmacy Practice: G. Parthasarathi, Karin Nyfort-Hansen and Milap Nahata, Universities Press.
5. Basic & Clinical Pharmacology, Bertram G. Katzung, MC Graw Hill.
6. Basic Principles of Clinical Research & Methodology: SK Gupta, ICMI.
7. The Merck Manual of Diagnosis and Therapy. Mark H. Beers and Robert Berkow. Published by: Merck & Co.Inc., USA.
8. Clinical Pharmacy & Therapeutics, Roger Walker; Churchill Livingstone Publication.
9. Textbook of Therapeutics, Drug & Disease Management: Richard A. Helms, David J., Quan, Eric T Herfindal, Lippincott Williams & Wilkins.
10. Principles of Internal Medicine Vol. I & II by Harrison.
11. Guide to Good Prescribing: A Practical manual, WHO Action programme on Essential Drugs Geneva.
12. Pharmacovigilance: Ron Mann, Elizabeth Andrews; John Wiley & Sons Ltd.
13. An Introduction to Pharmacovigilance: Patrick waller; Wiley Blackwell.
14. Drug Information Handbook, series, Lexi-comp.
15. Stockley's Drug Interactions. Pharmaceutical Press.
16. Pathophysiology & Therapeutics for Pharmacists A basic for clinical pharmacy practice; Russell J Greene. & Norman D Harris.
17. Contemporary perspectives on clinical pharmacotherapeutics : Kamlesh Kohli; Elsevier Private Limited.
18. Text Book of Pharmacovigilance: SK Guptal ; Jaypee, ICRI India.
19. Applied Therapeutics. The Clinical use of drugs: Mary Anne, Koda- Kimble. Lippincot-williams & Wilkins.

20. Pharmacotherapeutics – a pathophysiologic approach. Josheph T. Dipiro. MC grow hill. Medical

Traditional Medicine and Herbal Drugs Technology (PH431)

(Theory & Practical)

Credits: 3 (Theory)	Contact Hours per week: 3 (Theory)
: 2 (Practical)	: 3 (Practical)

Course Outcome:

The student will be able:

- To understand basic concept of Ayurveda and describe composition and methods of preparation of different ayurvedic formulations.
- To describe and compare pharmacognostic study of Indian traditional drugs.
- To understand and describe various concepts and components of plant tissue culture.
- To apply knowledge in quality control, standardization of crude drugs as per WHO guideline.
- To describe various natural allergens and photosensitizing agents.

Outline of the Course (Theory):

No.	Unit	Minimum No. of Contact Hours	Approx. weightage %
1.	The holistic concept of drug administration in traditional system of medicines	10	20
2.	Studies of traditional drugs	12	30
3.	Plant tissue culture	10	20
4.	WHO guideline for standardization of herbal drugs.	5	15
5.	Natural allergens and photosensitizing agent	5	10
6	Drug discovery from natural products.	3	5
Total		45	100

Detailed syllabus (Theory):

Sr. No.	Name of Chapter	Reference	Hrs. Required
1	The holistic concept of drug administration in traditional system of medicines, Introduction to ayurvedic preparation like Arishtas, Asavas, Gutikas, Tailas, Churnas, Lehyas & Bhasmas, Ghrita- their preparation and standardization.	6,10,11,14	10
2	Studies of traditional drugs, common vernacular names, botanical sources, chemical nature of chief Constituents, pharmacology, categories and common uses and market formulations of following drugs: Amala, Satavari, Musali, Damvel, Bhilama, Nagod, Chitrak, Apamarg, Gokhru, Bhringraj, Arjuna, Guggulu, Madhunashini, Shilajit, Piper, Kapikachhu, Galo, Vaj, Ashwagandha, Shankhapushpi, Brahmi	1,2,3,4,7,8, 9,12,13,15, 16,17,24	12
3	Plant Tissue Culture- Techniques of initiation and maintenance of various types of cultures. Immobilized cell culture and biotransformation studies including recent development in production of biological active constituents in static, suspension and hairy root cultures.	10,14,15,22	10

4	Introduction to standardization of herbal and ayurvedic medicines. WHO guideline for standardization of herbal drugs.	18,19,20, 21,23	5
5	Natural allergens and photosensitizing agents.	10,14,18,21	5
6	Drug discovery from natural products.	2,4,5,19,20	3

Detailed syllabus (Practical):

Sr. No	Aim of the Practical
1	Pharmacognostic study of Madhunashini/nagod/damvel.
2	Pharmacognostic study of Satavari / Musali.
3	Pharmacognostic study of Galo.
4	Pharmacognostic study of Brahmi.
5	Pharmacognostic study of Shankhpushpi.
6	Pharmacognostic study of Withania.
7	Isolation and estimation of Piperine in crude drug or formulation.
8	Isolation and estimation Curcuminoids in crude drug or formulation.
9	Morphology of traditional drugs.
10	Standardization of Ayurvedic formulation.
11	Standardization of herbal raw material or formulation as per WHO guideline

Recommended study materials:

1. Ayurvedic medicinal plants, Kapoor L. D., CRC press, 1st Indian Reprint, 2005.
2. Compendium of Indian Medicinal Plants Vol. I, II, III, IV by R.P. Rastogi and B.N. Mehrotra, CDRL Lucknow, 1993.
3. Cultivation and Utilization of Aromatic Plants. Edited by C.K. Atal and B.M. Kapoor R.R.L., Jammu- Tawi, 1982.
4. Herbal Drug Industry. Chief Editor R.D. Chaudhary, 1st Edition, 1996. Eastern Publishers, New Delhi.
5. Herbal drug technology, S. S. Agrawal and M. Paridhavi, Univeristies Press, 1st Edition, 2007.
6. Indian Herbal Pharmacopoeia, 1999. Vol. I and Vol. II. A Joint Publication of R.R.L. Jammu and IDMA, Mumbai.
7. Indian Materia Medica. Edited by K.M. Naadkarni. Volume Iand II. Reprint 1996. Bombay Popular Parkashan.
8. Kiritkar K.R. and Basu B.D. Indian Medicinal Plants. Text Volume I, II, III and IV 1987. International Book Distributors, Dehradun.
9. Pharmacognosy of Ayurvedic drugs, Kerala Series, volume 1 to 10.
10. Textbook of Pharmacognosy and Phytochemistry, Biren Shah and A K. Seth, Elsvier Publication, 1st Edition, 2010.
11. The ayurvedic Formulary of India, Part I and II, Department of Indian system of Medicine and Homoeopathy, Govt, of India, 1st English Edition, 2000.
12. The Useful Plants of India, 3rd Reprint, 1994. Pub. & Information Directorate CSIR, New Delhi.
13. The wealth of India. Raw Materials Revised. Edition 1985. Publication and Information Directorate, CSIR, New Delhi.
14. Trease and Evans Pharmacognosy, 15th Edition, 2008. W.B. Saunders Company, Singapore.

15. Vaidya B. G. Some Controversial Drugs in Indian Medicine. 1st Edition, 1982. Chaukhaba Orientalia, Varanasi.
16. Microscopic profile of powdered drugs used in Indian systems of medicine, Malti G. Chauhan and Pillai A.P.G., volume 1, Leaf drugs, (2005), Gujarat Ayurved University, Jamnagar.
17. Microscopic profile of powdered drugs used in Indian systems of medicine, Malti G. Chauhan and Pillai A.P.G., volume 1, Bark drugs, (2007), Gujarat Ayurved University, Jamnagar.
18. Pharmacognosy and pharmacobiotechnology, Ashutosh Kar, New Age International (P) Ltd, Publishers, 2nd edition 2007.
19. Quality Standards of Indian Medicinal Plants, Volume I and II, A. K. Gupta, ICMR, 2003.
20. Quality Control of Herbal Drugs: Mukherji P. K., Business Horizon Pharma. Publishers, 1st Edition, 2002.
21. Pharmacognosy: C.K.Kokate, A.P.Purohit, S.B.Gokhale, Nirali prakashan, Pune, 39th Edition, 2007.
22. Method in Plant tissue culture by U. Kumar Agro botanica 1999. Bikaner India.
23. Quality control methods for medicinal plant materials, W.H.O., Geneva, A.I.T.B.S. Publishers and distributors, 1st Indian Edition, 2002.
24. The Ayurvedic Pharmacopoeia of India, Government of India, Ministry of Health and Family Welfare, Department of ayurveda, yoga & naturopathy, Unani, siddha and homoeopathy,New delhi, 2007.

PHARMACEUTICAL ANALYSIS – III (PH432)

(Theory & Practical)

Credits : 3 (Theory) : 2 (Practical)	Contact Hours per week : 3 (Theory) : 3 (Practical)

Course Outcome:

The student will be able:

- To apply appropriate spectroscopic and thermal methods for analysis of pharmaceuticals based on basic understandings of technique.
- To describe instrumentation and operating principle of spectroscopic and thermal techniques.
- To interpret the spectroscopic data and establish the identity of the compound.
- To understand basic concepts and components of Quality Control, Quality Assurance, Good Manufacturing Practices and Good Laboratory Practices.
- To independently process, interpret the data obtained through experimentation and report the results.

Outline of the Course (Theory):

No.	Unit	Minimum No. of Contact Hours	Approx. weightage %
1	Spectroscopy	02	04
2	Ultraviolet/Visible Molecular Absorption Spectroscopy	08	18
3	Infrared Spectroscopy	05	11
4	Nuclear Magnetic Resonance Spectroscopy	07	16
5	Molecular Mass Spectrometry	05	11
6	Molecular Luminescence Spectroscopy	03	07
7	Raman Spectroscopy	02	04
8	Atomic Absorption and Atomic Emission Spectroscopy	03	07
9	X-ray methods	03	07
10	Thermal Methods of Analysis	05	11
11	Quality Assurance	02	04
	Total	45	100%

Detailed syllabus (Theory):

Sr. No.	Unit	Minimum No. of Contact Hours	Ref.
1	**Spectroscopy:** General Properties of Electro-magnetic Radiation (EMR), Interaction of EMR with matter, Classification of spectroscopic methods.	02	1,2,3
2	**Ultraviolet/Visible Molecular Absorption Spectroscopy:** UV/Vis absorption by molecules, Electronic transitions, Factors affecting absorption of radiation by molecules, Beer-Lambert's Law and its deviations, Instrumentation,	08	1,2,3

	Calibration of Spectrometer, Qualitative and quantitative applications, Recent advancements.		
3	**Infrared Spectroscopy:** Origin of IR spectra, Molecular vibrations, Fundamental bands, Vibrational frequencies, Fermi Resonance, Overtones, Instrumentation, Sample handling, Fourier transform infrared spectroscopy (FTIR), Felllgett's advantage, Recent advancements in sampling techniques and accessories, Applications and interpretation of IR spectrum.	05	2,4,5
4	**Nuclear Magnetic Resonance Spectroscopy:** NMR phenomenon, Theory, Origin of NMR spectrum, NMR spectrometers – instrumentation and sample handling, Chemical shift and its measurement, Shielding-deshielding, Factors affecting chemical shift, Spin-spin splitting, Coupling constant, NOE, Shift reagent, Interpretation of NMR spectrum and brief introduction to application of NMR. C^{13} NMR – Introduction, Principle and Applications.	07	2,3,4,5
5	**Molecular Mass Spectrometry:** Introduction, Principle, Origin of mass spectra, Instrumentation – mass spectrometers, Types of ions, Fragmentation pattern, Interpretation of mass spectra, Applications and recent advancement. Overview of hyphenated techniques like LC-MS and GC-MS.	05	2,4,5,6
6	**Molecular Luminescence Spectrometry**: Theory of Fluorescence and Phosphorescence, Factors affecting the intensity of chemiluminescence, Instrumentation, Analytical applications and recent advancement.	03	1,2
7	**Raman Spectroscopy**: Theory, Instrumentation and Applications	02	1,2
8	**Atomic Absorption and Atomic Emission Spectroscopy**: Introduction, Origin of atomic absorption & atomic emission spectra, Instrumentation, Applications. Flame photometry.	03	1,2
9	**X – Ray Methods**: Brag's law and its equation, X-ray absorption and X-ray diffraction methods, Applications.	03	1
10	**Thermal Methods of Analysis**: Introduction, Classification, Thermogravimetric analysis (TGA), Differential scanning calorimetry (DSC), Differential thermal analysis (DTA), etc., Their principle, instrumentation and applications	05	1,7
11	**Quality Assurance:** Introduction to concept of Quality Assurance, Good Manufacturing Practices and Good Laboratory Practices. ICH guidelines for quality.	02	12

Detailed syllabus (Practical):

Sr. No.	Aim of the Practical	Reference
1	A. List of Requirement.	8
	B. Standards of Tablets and Capsules as per Indian Pharmacopoeia.	
2	A. Calculations for weight variation, content of active ingredient and content uniformity test and dilution steps (step-up and step-down)	1,8
	B. Demonstration of UV Visible Spectrophotometer	

3	A. To determine wavelength maxima of given drug sample. B. To perform tests for weight variation and content of active ingredient for given Paracetamol tablets as per IP.	8,9
4	To perform weight variation of content of active ingredient tests for given Chloramphenical capsules IP.	8,9
5	To perform weight variation and content of active ingredient tests for given Atenolol tablets as per IP.	8,9
6	To perform the content uniformity test for given Riboflavin tablets as per IP.	8,9
7	A. To find out concentration of Na+ ion from given sample using flame photometer. B. To perform assay of Ringer's solution.	9
8	To perform the assay of quinine sulphate by fluorimeter using calibration curve method.	9
9	To study the effect of quenching on quinine sulphate by KI	9
10	To perform assay for Co-trimoxazole tablets as per IP.	8,9
11	Workshops on interpretation of given spectral data for structure elucidation	10,11
12	Any other experiments based on the topics given in theory course.	--

Recommended study materials:

1. Principles of Instrumental Analysis, Skoog, Hollar and Nieman, Saunders college Publishers, Philadelphia.
2. Pharmaceutical Analysis, 2nd Ed., David G. Watson, Elsevier Publication.
3. Pharmaceutical Analysis, Volume II Instrumental Methods, A.V. Kasture, K.R. Mahadik, S.G. Wadodkar, H.N. More, Nirali Prakashan, Pune, India.
4. Spectroscopic identification of organic compounds. R.M. Silverstein, G.C. Bassler, T.C. Morrill Pub: John Wiley and Sons, NY.
5. Organic Spectroscopy, W. Kemp, 3rd ed, ELBS publication, NY.
6. Spectroscopy, D.L. Pavia, G.M. Lampman, G.S. Kriz, J.R. Vyvyan, Cenage Learning, New Delhi, India.
7. Vogel's Textbook of Quantitative Chemical Analysis, J Mendham, R.C. Denney, J.D. Barnes, M. Thomas, B. Sivasankar, 6th Ed., Pearson Publication, New Delhi, India.
8. Indian Pharmacopoeia
9. Practical Pharmaceutical Chemistry, Part Two, A. H. Beckett & J. B. Stenlake, CBS Publication, India.
10. Applications of Absorption Spectroscopy of Organic compounds J. R. Dyer, Prentice Hall, London.
11. Organic Structures from Spectra, L.D. Field, S. Stemhell, J.R. Kalman, 5th Ed., Wiley Publication, UK.
12. Quality Assurance of Pharmaceuticals, Volume 1, World Health Organization. Geneva.

MEDICINAL CHEMISTRY-III (PH418)

(Theory)

Credits: 3	Contact Hours per week: 3

Course Outcome:

The student will be able:

- To understand the general synthetic pathways of drugs (Theoretically/Practically) it's therapeutics use, including its chemical classification, therapeutics uses and mechanism of action under the classes of chemotherapeutic agents and antibiotics.
- To understand structural activity relationship of medicinal class of drugs.
- To understand a novel appproach like" Microweave and Green Chemistry for synthesis of various chemical entities.

Outline of the course (Theory):

No.	Unit	Minimum No. of Contact hours.	Approx. Weightage %
1	The following classes of drugs will be discussed in relation to: Introduction to the rational and recent development of the drug (if any) including • Introduction • Chemical classification (if any) • Chemical nomenclature • Mechanism of action • Synthesis of the agents mentioned in bracket • Structure activity relationship • Therapeutic Uses **Chemotherapeutics** **Antibiotics**	42	93
2	Green Chemistry and its twelve principle and application & Microwave synthesis (Principle, application and advantages)	03	07
Total		**45**	**100 %**

Detailed syllabus (Theory):

No.	Unit	Minimum No. of Contact Hours	Reference
1	The following classes of drugs will be discussed in relation to: Introduction to the rational and recent development of the drug (if any) including • Introduction • Chemical classification (if any) • Chemical nomenclature • Mechanism of action • Synthesis of the agents mentioned in the bracket • Structure activity relationship • Therapeutic Uses		1,2,3,4,5,6
1.1	Chemotherapeutics • Sulphonamides and fluoroquinolones (sulphanilamide, sulphacetamide, sulphaguanidine, sulphathiazole, sulphadiazine, sulphafurazole, sulphamerizine, sulphamethoxazole) • Antimalarials (chloroquin, primaquin, amodiaquin, mepacrin hydrochloride, pyrimethamine) • Antimycobacterials (Antileprotic & Antitubercle agents) (isonizid, para amino salicylic acid, pyrazinamide, ethambutol, ethionamide, prothionamide, meprazinamide) • Antifungal agents (metronidazole, fluconazole) • Antiviral drugs including Anti-HIV drugs (amantadine) • Antineoplastic agents (methotrexate, chlorambucil, mustine, thio TEPA, cyclophospomide, 6-merceptopurine, hydroxylurea) • Immunosuppressive agents • Antiamoebic and Anthelmentic agents (piperazine, diethyl carbamazine citrate, thiabendazole, pyrantel, niclosamide) • Antiseptics and Disinfectants	27	1,2,3,4,5,6
1.2	**Antibiotics:** β-lactams, aminoglycosides, tetracyclines, macrolides, polyene & polypeptide antibiotics, chloramphenicol. (ampicillin carbenicillin, cephalexin, penicillin-V, chloramphenicol)	15	1,2,3,4,5,6
2	Green Chemistry and its twelve principle and application & Microwave synthesis (Principle, application and advantages)	03	7,8

Recommended Study Material:

1. Wilson and Giswold's Textbook of Organic, Medicinal and Pharmaceutical Chemistry,J. N. Delagado and W. A. R. Remers, Eds, J. Lipponcott Co. Philadephia.
2. Principles of Medicinal Chemistry by W. C. Foye,, Lea & Febiger, Philadelphia.
3. Burger's Medicinal Chemistry, H. E. Wolff, Ed. John Wiley & Sons, New York Oxford University Press, Oxford.
4. Instant Notes: Medicinal Chemistry, 2nd Edition, Graham L. Patrick by Taylor & Francis
5. Medicinal Chemistry: An Introduction by Dr. Gareth Thomas, John Wiley & Sons, 2008
6. Text book of Medicinal Chemistry by S. Alagarsamy, Vol. I & II, Elsevier Publication.
7. Practical Microwave Synthesis for Organic Chemists: Stretegies, Instruments and Protocols, 1st Edition, Wiley- VCH.
8. Green Chemistry and Engineering: A Practical Design Approach by David J. C. Constable 1st Edition, Wiley Publication.
9. Strategies for Organic Drug Synthesis & Design by Daniel Lednicer, John Wiley & sons, USA.
10. Organic Chemistry by I. L. Finar, Vol. I & II, ELBS/ Longman, London.

PHARMACEUTICAL TECHNOLOGY-II (PH433)

(Theory & Practical)

Credits: 3 (Theory) : 2 (Practical)	Contact Hours per week: 3 (Theory) : 3 (Practical)

Course Outcome:

The student will be able:

- To employ the knowledge of pharmaceutical technology used in production of sterile dosage forms, liquid dosage forms, aerosol and cosmetic products with good manufacturing practices.
- To describe in detail instrumentation/equipments for formulation and evaluation of pharmaceutical products as per standards.
- To explain the fundamentals of pilot plant scale up techniques.

Outline of course:

No.	Unit	Minimum No. of Contact Hours Approx.	Weightage %
1	Sterile Dosage Forms	13	27
2	Pharmaceutical Aerosols	10	23
3	Cosmetology and Cosmetic preparation	7	17
4	Liquid Dosage Forms	11	27
5	Pilot Plant Scale up: An overview	4	06
6	Total	45	100

Detail syllabus (Theory):

1	**Sterile Dosage Forms:** a) Design and requirements for sterile products manufacturing facility: i) Environmental controls: quality of air, HEPA filters and laminar flow, class 100,1000,10000 area, monitoring of Areas ii) Requirements of Aseptic Area for LVP/SVP as per GMP b) Flow plan for manufacturing of sterile products by terminal sterilization, Aseptic Process c) Introduction to form fill seal (FFS) technology. d) Sterile Products for Injection: i. Large volume parenterals (LVP) ii. Small Volume parenterals (SVP)	1,2,3,4	13

	iii. Sterile Suspension iv. Sterile emulsion v. Sterile dry powders vi. Sterile solutions e) Containers & closures for sterile products f) Evaluation of sterile products g) Sterile products for ophthalmic Application: Introduction, h) Ophthalmic Products: classification, formulation and preparation of eye drops, ophthalmic suspension, eye lotion, ophthalmic ointments, contact lens solutions. Solid inserts for ocular drug delivery (erodable & non erodable), in situ gel, and advancement in ophthalmic delivery system including intra ocular dosage forms; containers and closures.		
2	**Pharmaceutical Aerosols:** Definition, Advantages, disadvantages, Design of Aerosol: liquefied gas system, compressed gas system, barrier packs, components of aerosol, propellants: properties, classification, container for aerosol, valve assembly and components of valve, manufacturing: cold and pressure filling technique, Quality control tests, stability, physicochemical & biological evaluation. Dry powder and meter dose valve, meter dose inhaler; Recent development in MDI/DPI	1,2,5,6	10
3	**Cosmetology and Cosmetic preparation:** Fundamentals of cosmetic science, structure and function of skin and hair. Formulation, preparation and packaging of products such as Nail Preparation, Dental Products, Lipstick, Skin care products, Shampoo, shaving cream, sunscreen lotion and herbal cosmetic formulation.	10,11,12	7
4	**Liquid Dosage Forms:** Solution: methods of enhancing solubility, vehicles, A. Formulation components: additives (solubilizers, antioxidants, sweeteners, preservatives, colors, flavors.	1,3,7,8,9,13	11

	Manufacturing and filling methods, evaluation of liquids B. Suspension: definition, ideal requirements, formulation (Vehicle, suspending agents, antioxidants, preservatives, flocculating and deflocculating agents, wetting agents), approaches to formulation, manufacturing: equipments and process, flow sheets, Evaluation and stability of suspension. C. Emulsion: Definition, emulsion types, formulation: choice of aqueous phase and oil phase, choice of emulsifiers, other additives (antioxidants, preservatives, colors, flavors, sweeteners), manufacturing: equipments and process, flow sheets. Evaluation and stability of emulsion		
5	Pilot Plant Scale up: An overview	1	4

Detailed syllabus (Practicals):

Sr. No.	Aim of Practical
1	A. Demonstration of ampoules filling and sealing machine B. Preparation of emulsion by different methods
2	A. Preparation and evaluation Dextrose Injection. B. Demonstration of Bottle Sealing Machine.
3	A. Preparation and evaluation Ascorbic acid Injection. B. Preparation and evaluation of paracetamol suspension.
4	A. Preparation and evaluation Calcium Gluconate Injection. B. Demonstration of Aerosols/ MDI/DPI.
5	Preparation and evaluation Caffeine citrate injection.
6	Preparation and evaluation diclofenac sodium injection
7	Preparation of emulsion and to evaluate different parameters of quality of emulsion
8	Preparation of A. Vanishing Cream B. Clear Shampoo C. Cream Shampoo
9	Preparation of A. Face Powder B. Lather Shaving Cream C. Foam Shaving Cream
10	Preparation of A. Lipstick B. Nail Polish C. Nail Polish Remover
11	Preparation of A. Tooth Powder

	B. Tooth Paste
12	A. To Prepare emulsion and find out the type of emulsion by measuring different evaluation parameter. B. Preparation of eye drops & eye ointment
13	Preparation of A. Cleansing Cream B. Cleansing Lotion C. Cold Cream

Recommended study materials:

1. The Theory & Practice of industrial pharmacy, Leon Lachman, Lea & Febiger, Varghese Publication House Bombay.
2. Remington's Pharmaceutical Sciences , Edited by A. R. Gennaro, Mack Publishing Co,
3. Modern Pharmaceutics, Edited by G. S. Banker & C.T. Rhodes, Marcel Dekker inc. N.Y.
4. Pharmaceutical dosage forms: Parenteral, Vol.: 1, 2, 3, Leon, Lachman, K. E. Avis, Marcel Dekker inc. N.Y.
5. Pharmaceutics: The science of dosage form design, Edited by M.E. Aulton. Churchill Livingstone, New York.
6. Ansel's Introduction to Pharmaceutical Dosage Form and Drug Delivry system, H.C. Ansel, N.G. Popovich, Lippincott Williams and wilkins, Philadelphia.
7. Indian Pharmacopoeia , published by Indian Pharmacopoeial commission, Ghaziabad
8. United State Pharmacopoeia, Indian edition, United State Pharmacopoeial convention INC.
9. British Pharmacopoeia, British Pharmacopoeia commission office, U.K.
10. Cosmetic- Formulation, Manufacturing and Quality control, P. P. Sharma, Vandana Publication Pvt. Ltd., New Delhi.
11. Cosmetic Science and technology, Vol-1-3, M. S. Blsam, Wiley India Pvt. Ltd.
12. Poucher's Perfumes, Cosmetics and Soaps, Edited by Hilda Butler, Springer, U.K.
13. Pharmaceutical dosage forms: Disperse systems, Vol.: 1, 2, 3, H.A. Lieberman, G. S. Banker , Marcel Dekker inc. N.Y.

Note: It is preferable to refer latest addition of all above mentioned books/ Pharmacopeia

Pharmacy Practice -II (PH420.1)

Credits: 2 (Theory) Contact Hours per week: 2 (Theory)

Course Outcome:

The student will be able:

- To learn the concept of health economics.
- To understand the role of the pharmacist in Ethics and roles in Ethical issues in pharmacy practice research.
- To develop skills of research in clinical pharmacy practice.

Outline of course:

No.	Unit	Minimum No. of Contact Hours Approx.	Weight age %
1	Role of Pharmacist in Healthcare research, policy and economics	08	25
2	Ethical issues in clinical pharmacy practice research	08	25
3	Research in clinical pharmacy practice	14	50
	Total	**30**	**100%**

Detailed Syllabus

Sr no	Unit Details	No. of Hours
1	Role of Pharmacist in Healthcare research, policy and economics: Pharmacist or clinical pharmacist involvement in framing health care policy in various countries	08
2	Ethical issues in clinical pharmacy practice research:History, Codes, declarations and guidelines, basic principle of research, roles and responsibilities of ethics committee	08
3	Research in clinical pharmacy practice:Types of research, role of Pharmacist in clinical practice research	14

References:

1. A Textbook of Clinical Pharmacy Practice: G. Parthasarathi, Karin Nyfort-Hansen and Milap Nahata, Universities Press.
2. Practice of Hospital Clinical and community pharmacy, By Mohd Aqil.
3. Indian Guidelines for Ethical conduct of research in Human Subjects
4. Developing Pharamcy Practice: A focus on patient care. Handbook 2006 edition.
5. Research, review and other literature from internet sources

HERBAL FORMULATIONS-II (PH 420.2)

Credits: 2 Contact Hours per week: 2

Course Outcome:

The student will be able:

- To summarize different types preformulation studies and their significance in formulation designing.
- To develop formula of the selected formulations containing herbal extracts with scientific justifications.
- To decide the quality attributes to be assessed for selected formulations.

Outline of the Course:

No.	Unit	Minimum No. of Contact Hours	Approx. weightage %
1	Preformulation and Formulation Design	5	25
2	Preparation of Herbal Therapeutic Formulations	15	40
3	Herbal Cosmetics and Nutraceuticals	10	35
	Total	**30**	**100**

Detailed Syllabus:

No.	Unit details	Contact Hours	References
1	Concept of preformulation studies and fundamentals of formulation design.	5	1,2,3,4,6
2	Herbal formulations- topical, tablets, syrups, dentifrices	15	2,3,4,5,6
3	Herbal cosmetics- skin care and hair care.	10	5,6

Recommended study materials:

1. Harry's Cosmeticology, Radelph Harry, 8th edition, Chemical Publishing Company.
2. Perfumes, Soaps, Detergents and Cosmetics, S.C. Bhatia, 1st edition, CBS publishers
3. Handbook of Cosmetic Science and Technology, Andre Barel, Marc Paye, Howard I. Maibach, CRC Press.
4. Herbal Drug Technology, S. S. Agrawal and M. Paridhavi, Univeristies Press, 1st Edition, 2007.
5. Quality Control of Herbal Drugs: Mukherji P. K., Business Horizon Pharma. Publishers, 1st Edition, 2002.
6. Indian Herbal Pharmacopoeia, Volume I and II, 1999. A Joint Publication of IDMA and RRL, Jammu –Tawi.

Production of Bulk Drugs and Drug Intermediates II (PH420.3)

Credits: 2 (Theory) Contact Hours per week: 2 (Theory)

Course Outcome:

The student will be able:

- To explain basic concept of Process Chemistry along with Unit process for buld drug manufacturing.
- To demonstrate various techniques and steps involve in a buld drug manufacturing in pharmaceutical industries.
- To understand various environmentally safe synthetic pathway for synthesis of drug intermediates.
- To describe the process for product registration.

Outline of the Course:

No.	Unit	Minimum Contact Hours	Approx. weightage %
1	Understanding of concepts of process chemistry with references to various unit processes	30	100
Total		**30**	**100**

Detailed Syllabus (Theory):

No.	Unit details	Contact Hours	References
1	**Understanding of concepts of process chemistry with references to various unit processes** • Route Enabling & Lab Optimization • Kinetics & Mechanism evaluation in PRD • Optimization by understanding impurities • New synthesis technologies like solid phase chemistry, flow chemistry, micro wave and aqua Chemistry • Advanced Asymmetric Synthesis & Catalysis • Knowledge of the concepts of asymmetric synthesis & applications • Applications of enzymes in organic synthesis • Understanding the above mentioned concepts with reference to unit process like: 1. Alkylation 2. Amination 3. Condensation and Addition 4. Esterification 5. Friedel-Crafts Reactions 6. Halogenations 7. Hydrogenation 8. Oxidation	30	1,2,3,4,5

Recommended study materials:

1. Process Chemistry in the Pharmaceutical Industry, Volume Iedited by Kumar Gadamasetti

2. N. G. Anderson, ''Practical Process Research & Development'', Academic Press 2000.
3. O. Repic, ''Principles of Process Research and Chemical Development in the Pharmaceutical Industry, Wiley 1998.
4. N. Yasuda, “The Art of Process Chemistry”, Wiley 2010
5. R.A. Sheldon, I. Arends, U. Hanefeld, ''Green Chemistry & Catalysis'', Wiley 2007

Pharmaceutical Quality Assurance and Quality Control – II (PH420.4)

Credits: 2 (Theory) Contact Hours per week: 2 (Theory)

Course Outcome:

The student will be able:

- To elaborate the role of validation in assurance of quality of pharmaceutical industries.
- To describe various aspects of documentation – SOPs, records and reports required for pharmaceutical Industries.
- To explain the types, approaches and reports of quality inspections and audits.

Outline Syllabus

No.	Unit	Minimum Contact Hours	Approx. weightage %
1	Qualification and Validation	09	30
2	Analytical Method Validation	03	10
3	Documentation in Pharmaceutical Industry	09	30
4	Quality Inspections and Audits	09	30
Total		30	100

Detailed Syllabus (Theory):

No.	Unit	Minimum Contact Hours	References
1	Qualification and Validation: Definitions, Difference between Qualification and Validation. Introduction to the basic concept of Validation in Pharmaceutical Manufacturing, Process Validation: Types, Objective, how it differs from Qualification (DQ, IQ, OQ, PQ), Validation Master Plan	09	
2	Analytical Method Validation: Objectives, Method Characteristics, Assessment Methodology	03	
3	Documentation in Pharmaceutical Industry: Good Documentation Practices, Batch Formula Record, Master Formula Record, Quality Audit Report, SOP, Certificate of Analysis, Complaint and evaluation of complaint, Handling of returned goods, Recalling and Waste Disposal.	09	
4	Quality Inspections and Audits: Definitions- Inspection, Periodic evaluation and Audit, Types, Approaches, Benefits, Audit Report, Regulatory Audit	09	
Total		30	

Recommended study materials:

1. Pharmaceutical Process Validation, B. T. Loftus & R. A. Nash, Drugs and Pharm Sci. Series, Vol. 129, Marcel Dekker Inc., New York.
2. The Theory & Practice of Industrial Pharmacy, Leon Lachman, Herbert A. Lieberman, Joseph. L. Karig, Varghese Publishing House, Bombay.
3. Validation Master Plan, Terveeks, Davis Harwood International publishing.
4. Validation of Aseptic Pharmaceutical Processes, Carleton & Agalloco, Marcel Dekker Inc., New York.

5. Pharmaceutical Process Scale-Up", Michael Levin, Drugs and Pharm. Sci. Series, Vol. 157, Marcel Dekker Inc., New York.
6. Lachman L Liberman Theory and practice of industrial pharmacy by 3 rd edition
7. Sidney H Willing, Murray M, Tuckerman. Williams Hitchings IV, Good manufacturing of pharmaceuticals (A Plan for total quality control) 3rd Edition. Bhalani publishing house Mumbai.
8. Tablets Vol. I, II, III by Leon Lachman, Herbert A. Liberman, Joseph B. Schwartz, 2nd Edn.(1989) Marcel Dekker Inc. New York.
9. Text book of Bio- Pharmaceutics and clinical Pharmacokinetics by Milo Gibaldi, 3rd Edn, Lea & Febriger, Philadelphia.
10. Pharmaceutical process validation (Drugs and Pharmaceuticals Series), Ira R. Berry and Robert A. Nash, 2nd Edn.(1993), Marcel Dekker Inc., New York.
11. Good laboratory Practice Regulations – Allen F. Hirsch, Volume 38, Marcel Dekker Series, 1989.

DRUG REGULATORY AFFAIRS- II (PH420.5)

Credits: 2 (Theory) Contact Hours per week: 2 (Theory)

Course Outcome:

The student will be able:

- To understand the importance of WHO guidelines for the drugs and Pharmaceuticals relevant to the regulatory filing.
- To understand the importance of guidelines of International Conference on Harmonization (ICH) relevant to the regulatory filing.
- To prepare a dossier as per the guidelines of ASEAN countries.

Outline of course:

No.	Unit	Minimum No. of Contact Hours Approx.	Weight age %
1	International Conference on Harmonization (ICH) guidelines on stability relevant to the regulatory filing	14	47
2	WHO guidelines for the drugs and Pharmaceuticals	08	27
3	ASEAN guidelines: Study and dossier preparation as per the guidelines of ASEAN	08	26
	Total	**30**	**100%**

Detailed Syllabus

Sr no	Unit Details	No. of Hours
1	International Conference on Harmonization (ICH) guidelines on stability relevant to the regulatory filing: Stability Testing of New Drug Substances and Products Stability Testing , Photostability Testing of New Drug Substances and Products, Stability Testing for New Dosage Forms, Bracketing and Matrixing Designs for Stability Testing of New Drug Substances and Products, Evaluation of Stability Data, Impurities in New Drug Substances, New Drug Products	14
2	WHO guidelines for the finished products	08
3	ASEAN guidelines: Study and dossier preparation as per the guidelines of ASEAN	08

Pedagogy:

- Classroom teaching :- 15 hrs
- Case study :- 05 hrs
- Group Discussions :- 05 hrs
- Students' Presentations :- 05 hrs

References:

1. Medical Product Regulatory Affairs: Pharmaceuticals, Diagnostics, Medical Devices By John J. Tobin and Gary Walsh

2. FDA Regulatory Affairs: A Guide for Prescription Drugs, Medical Devices, and Biologics, Second Edition by Douglas J. Pisano and David S. Mantus
3. Good Drug Regulatory Practices: A Regulatory Affairs Quality Manual (Good Drug Development Series, Volume 1 by Helene I. Dumitriu
4. Encyclopedia of Pharmaceutical Technology, James Swarbrick and James C. Boylan, Marcel Dekker Inc., New York.
5. Guidelines for Developing National Drug Policies; WHO Publications, 1998.
6. **www.ich.org**
7. www.**who.int**/
8. www.**asean**.org/

Sterile Products Technology –II (PH420.6)

Credits: 2 (Theory) Contact Hours per week: 2 (Theory)

Course Outcome:

The student will be able:

- To understand the various routes of administration and the other anatomical aspects for the administration of sterile dosage forms.
- To understand applications of sterile dosage forms in manufacturing and packaging of drugs/dosage forms.
- To get familiar with various sterile dosage forms and its applications.

Outline of course:

No.	Unit	Minimum No. of Contact Hours Approx.	Weightage %
1	Introduction to Sterile dosage forms	08	27
2	Formulation of Sterile dosage forms	12	40
3	Quality Control of Sterile Products	10	33
	Total	**30**	**100%**

Detailed Syllabus

Sr. no.	Unit Details	No. of Hours
1	Introduction to Sterile dosage forms: • Introduction of Parenteral and non-parenteral dosage forms. • Approaches for drug delivery and associated Complications. • Application of sterile dosage forms.	08
2	Formulation of Sterile dosage forms: • Formulation methods, Manufacturing and packaging of Parenterals with focus on equipment for Large volume parenterals (LVP), Small Volume parenterals (SVP), Suspension, Emulsion, Dry powders, Solutions etc. • Formulation methods, manufacturing & packaging of ophthalmic with focus on equipment for Eye drops, ophthalmic suspension, eye lotion, ophthalmic ointments, contact lens solutions etc.	12
3	Quality Control of Sterile Products: Evaluation of sterile products with respect to sterility and Package integrity.	10

Recommended Study Materials:

1. The Theory & Practice of industrial pharmacy, Leon Lachman, Lea & Febiger, Varghese Publication House Bombay.
2. Pharmaceutical dosage forms: Parenteral, Vol.: 1, 2, 3, Leon, Lachman, K. E. Avis, Marcel Dekker inc. N.Y.
3. Pharmaceutics: The science of dosage form design, Edited by M.E. Aulton. Churchill Livingstone, New York.

4. Handbook of Pharmaceutical Manufacturing Formulations: Sterile Products, Sarfaraz K. Niazi Volume 6, Informa Healthcare USA, Inc.2009.
5. Parenteral Quality Control: Sterility, Pyrogen, Particulate, and Package Integrity Testing, Michael J. Akers, Daniel S. Larrimore, and Dana Morton Guazz, volume 125.

ENVIRONMENTAL SCIENCES (PH421)

(Theory)

Credits: 2	Transactional Hours/Week: 2 (Theory)

Course Outcome:

The student will be able:

- To understand the natural environment and its relationships with human activities.
- To explain the methods for conservation of natural resources.
- To integrate the facts, concepts and methods to environmental issues in pharmaceutical industry.

Outline of the Course:

Sr No.	Unit
1	Environmental Science
2	Ecology and Ecosystems
3	Biodiversity and its conservation
4	Natural Resources and their conservation
5	Environmental Pollution
6	Field Work
7	Environmental issues related to the specific discipline for Pharmacy Course

Topic No.	Name of topic and contents	Hrs.
1.	**Environmental Science:** Definition, scope and importance. Need for public awareness.	02
2.	Ecology and Ecosystems. Definition of ecology, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem, Food chains, food webs and ecological pyramids. Ecosystem: Forest ecosystem, Grassland ecosystem, Desert Ecosystem, Aquatic ecosystem (ponds, streams, lakes, rivers, oceans, estuaries)	04
3.	Biodiversity and its conservation. Introduction – Definition: genetic, species and ecosystem diversity. Value of biodiversity: consumptive use, productive use, social, ethical, and aesthetic and option values. Biodiversity at global, National and local levels. Regulations in India for Biodiversity Hot Spots of biodiversity. Threats to biodiversity: habitat, poaching of wildlife, man wildlife conflicts. Endangered and endemic species of India.	04
4.	Natural Resources and their conservation. **Air resources:** Composition, structure, air quality management. **Forest resources:** Use and over/exploitation, deforestation, case studies, timber extraction, mining, dams and their effect on forests and tribal people. • **Water resources:** Use and over –utilization of surface and ground	06

	water, flood, drought, conflicts over water, dams –benefits and problems; water quality management: management of water resources. E.g. rivers, lakes, ground water, etc: fluorosis and arsenic problems. • **Mineral resource:** draw on and exploitation, environmental effects of extracting and using mineral resources, case studies. • **Food resources:** World food problems, changes caused by agricultural and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. • **Energy resources:** Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources, Case studies.	
5.	Environmental Pollution. **Air pollution**: Definition, causes, effects and control measures: Air Quality Management, Carbon credit, Air Pollution Case Studies. **Water pollution**: Definition, causes, effects and control measures: case studies: Water Marine Pollution Thermal pollution. **Soil pollution**: Definition, causes and control measures: case studies Noise pollution. **Nuclear hazards**: Waste Management: **Quality Management**: Definition, causes, effects and control measures. Waste minimization through cleaner technologies; reuse and recycling of wastes. Solid waste management Causes, effects and control measures of urban and industrial wastes: hazardous waste: bio medical waste. Pollution case studies. Disaster management: floods, earthquake, cyclone and landslide	06
6	Field Work Visit to local area to document environmental assets-river/ forest/ Grasslands/ Hill/ Mountain, Visit to a local pollution site-Urban/ Rural/ Industrial/Agricultures/etc. Study of common plants, insects, birds. Study of simple ecosystems- pond, river, hill slopes, etc.	05
7.	Environmental issues related to the specific discipline for Pharmacy Course. • Maintenance of healthy environment in Pharmaceutical industry • Disposal of wastes, • Hospital waste, Pharmaceutical industrial waste. • Air sampling and air handling in Pharma. Industries.	03

Pedagogy

Sr. No.	Component	Approximate Sessions allotted
1	Classroom Contact Sessions	25 sessions of one hour
2	Field work	05 hour

Evaluation

The students' performance in the course will be evaluated on a continuous basis through following components

Internal Evaluation

Sr.No	Component	Number of such Exam	Marks per Incident	Total	Percentage of Total Internal
1	Case Study/Presentation	1	30	100	30
2	Written Test (MCQ Based)	1	50		50
3	Field work	1	20		20

Recommended Books:

A. Environmental Studies by Erach Bharucha.
B. Environmental Biology by P.C. Das, 1st Edition, ISBN: 9374734710/9789374734711, All India Travellers Booksellers Publishers & Distributors AITBS, 2011.
C. Shardha sinha, Manisha shukla, Arif siddiqui and Neeraj agarwal., 2008. A Textbook of Environment and Ecology for pharmacy Students, published by A.I.T.B.S. publishers, Delhi.
D. Shradha sinha, A text book of environmental sciences, 1st edition, All India Travellers Booksellers Publishers & Distributors AITBS, Delhi.

SYLLABI

(Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CLINICAL PHARMACY & PHARMACOTHERAPEUTICS-II (PH434)

(Theory & Practical)

Credits: 3 (Theory) Contact Hours per week: 3 (Theory)
: 2 (Practical) : 3 (Practical)

Course Outcome:

The student will be able:

- To counsel the patient in terms of patient compliance and appropriate drug use.
- To discuss the concept of rational prescribing & essential drug.
- To describe the theoretical considerations and basic principles of pharmacotherapeutics in the management of the diseases and conditions of various body system such as CNs, mascular system, endocrine , infectious and respiratory system with the help of standard treatment guidelines and treatment algorithms..
- To discuss the importance of individualized therapeutic plans based on patients condition.

Outline of the Course (Theory):

Sr. No.	Unit	Minimum No. of Contact Hours	Approx. Weightage %
1	Patient counseling	2	5
2	Concept of essential drugs and Rational use of medicines.	2	5
3	Pharmacoeconomics	3	6
4	Drugs used in special population	5	12
5	Introduction to Medication Error	2	5
6	Introduction to Clinical Research	2	5
7	Clinical Presentation, Diagnosis, Complication & Management of following Diseases/Disorders: a) CNS Disorders	6	12
	b) Endocrine Disorders	6	11
	c) Musculoskeletal Disorders	3	6
	d) Infectious Diseases	8	18
	e) Gastrointestinal Disorders	3	9
	f) Respiratory Disease:	3	6
Total		45	100

Detailed syllabus (theory):

Sr. No.	Name of Chapter	References	Hrs required
1	Patient counseling (to improve patient compliance on appropriate drug use)	4,8	2
2	Concept of essential drugs and Rational use of medicines.	4,8	2
3	Pharmacoeconomics: Cost benefit analysis, cost effectiveness analysis, cost minimization analysis, cost utility analysis	8	3

4	Drugs used in special population: Pediatric, Geriatric and pregnant women.	1,6,8,9	5
5	Introduction to Medication Error: Types and its detail with suitable example.	8,12	2
6	Introduction to Clinical Research: - Types of clinical research - Phases of clinical research	11,8	2
7	a) CNS Disorders: Epilepsy, Parkinsonism, Schizophrenia, Depression, Insomnia, Anxiety, Migraine etc	1,2,4,6,7,9	6
	b) Endocrine Disorders: Diabetes Mellitus , Thyroid disorders, Addison's disease	1,2,4,6,7,12	6
	c) Musculoskeletal Disorders: Rheumatoid arthritis, Gout & Hyperuricemia.	1,2,4,6,7,9	3
	d) Infectious Diseases: Tuberculosis, Urinary Tract Infections, Enteric infections, upper respiratory tract infections, Meningitis, Malaria, sexually transmitted diseases(STDs) and AIDS,	1,2,4,6,7, 9,12	8
	e) Gastrointestinal Disorders: Peptic ulcer disease, Inflammatory Bowel Disease, Hepatitis	1,2,4,6,7,9,12	3
	f) Respiratory Disease: Asthma & COPD	1,2,4,6,7,9,12	3

Detailed syllabus (Practical):

1	Prescription monitoring related to Polypharmacy and for rational drug therapy
2	Drug formulation (e.g. Cough & cold, Anemia & Diarrhea) Audit for rationality
3	Pharmacoeconomics related practical.
4	Case studies related to Pediatrics, Geriatrics & pregnancy.
5	Case studies based on CNS, Endocrine, Musculoskeletal, Infectious and Gastrointestinal Disease, Respiratory Disease.

Recommended study materials :(Latest edition)

1. Textbook of Therapeutics, Drug & Disease Management: Richard A. Helms, David J., Quan, Eric T Herfindal, Lippincott Williams & Wilkins.
2. Davidson's Principles & Practice of Medicines:Churchill Livingstone Elsevier.
3. Pathology and Therapeutics for pharmacists: A basis for clinical pharmacy practice, Greene and Norman, Pharmaceutical press.
4. Basic & Clinical Pharmacology,Bertram G. Katzung, MC Graw Hill.
5. Foundations of Clinical Drug therapy:Anne Collins Abrams, Lippincott Williams & Wilkins.
6. Clinical Pharmacy & Therapeutics, Roger Walker; Churchill Livingstone Publication.
7. Principles of Internal Medicine Vol. I & II By Harrison.

8. A Textbook of Clinical Pharmacy Practice: G. Parthasarathi, Karin Nyfort-Hansen and Milap Nahata, Universities Press, 2004
9. Elements of Clinical Pharmacy:R.K Goyal, Parloop A.Bhatt & Mahesh D.Burande, B.S. Shah Prakashan.
10. Handbook of Drug Interactions, A clinical & forensic Guide by Ashraf Mozayani: Humana Press.
11. Basic Principles of Clinical Research & Methodology: SK Gupta, ICMI.
12. The Merck Manual of Diagnosis and Therapy. Edited by Mark H. Beers and Robert Berkow. Published by: Merck & Co.Inc., USA.
13. Applied Therapeutics. The Clinical use of drugs: Mary Anne, Koda- Kimble. Lippincot-williams & Wilkins.
14. Pharmacotherapeutics – a pathophysiologic approach. Josheph T. Dipiro. MC grow hill. Medical
15. Drug Information Handbook, series, Lexi-comp.

MEDICINAL CHEMISTRY-IV(PH435)

(Theory)

Credits: 3(Theory)	Contact Hours per week: 3(Theory)
:1 (Practical)	:3 (Practical)

Course Outcome:

The student will be able:

- To understand the physicochemical properties of drugs and Structure Activity Relationship (SAR) in relation to drug design and discovery.
- To describe their knowledge and understanding of computational and conventional techniques applicable to drug discovery.
- To have knowledge and understanding of combinatorial chemistry techniques applicable to drug design and discovery.
- To apply earned theoritical knowledge to their experimental aspects for developing of drug candidate.

Outline of the Course (Theory):

No.	Unit	Minimum No. of Contact hours.	Approx. Weightage %
1	Strategies in the search for new lead compounds	08	20
2	QSAR and Ligand Based Drug Design	12	25
3	Structure Based Drug Design	12	25
4	Combinatorial Chemistry	08	20
5	Introduction to recent advances in drug design	05	10
Total		45	100 %

Detailed syllabus (Theory):

No	Unit	Minimum No. of Contact Hours	References
1	Strategies in the search for new lead compounds: introduction, improvement of existing drugs, systematic screening including extensive screening, random screening and high-throughput screening, screening of synthetic intermediates, selective optimization of side activities (SOSA) approach, new use for old drugs – an illustrative study with suitable examples.	08	1,2,3 4,5,6,7,8,9, 10
2	QSAR and Ligand Based Drug Design: introduction, SAR versus QSAR, various QSAR methods, linear regression and multiple linear regression analysis, Hansch analysis, Free-Wilson analysis, mixed approach, parameters used in QSAR, experimental and theoretical approaches for the determination of physicochemical parameters, statistical significance and interpretation of QSAR models, prediction of novel potent molecule, 2D QSAR, 3D QSAR-examples, CoMFA and CoMSIA, Pharmacophore generation and identification.	12	1,2,3 4,5,6,7,8,9, 10,11

3	Structure Based Drug Design: structure based drug design, requirement of SBDD, understanding of drug-receptor/enzyme/target interactions, preparation of protein/target along with active site analysis, docking process, various docking methods, De-novo drug design.	12	1,2,3 4,5,6,7,8,9, 10,11
4	Combinatorial Chemistry: introduction & principles of combinatorial chemistry and its importance in new drug discovery, various synthetic approaches and library Purification methods- solid support methods, encoding methods, combinatorial synthesis in solution, deconvolution and high throughput screening and its application to combinatorial library screening.	08	3,5,8,9,10
5	Introduction to recent advances in drug design: quantitative structure pharmacokinetic relationship (QSPR), bioinformatics, genomic & proteomics, Drug receptor interactions and drug metabolism in drug discovery.	05	1,2,3,7,11

Detailed Syllabus (Practical):

Topic No.	Aim of the Practical	References
1	Introduction to the apparatus used in organic synthesis involving complex assemblies like joints, flasks, condensers, guard tubes etc.	13, 14, 15
2	Importance of various experimental techniques used for organic synthesis like heating and cooling techniques, stirring and agitation with mechanical and magnetic stirrers, filtration, re-crystallization, extraction, distillation, reactions carried out under pressure and inert gas conditions, TLC and column chromatography.	13, 14, 15
3	Derivatization of some important organic chemicals and their identification by TLC and spectral data.	13, 14, 15
5	Case studies related to 2D and 3D QSAR.	9, 11, 16
6	Generation of QSAR equation from given data- its interpretation and its use in prediction of biological activity of novel compounds of a series.	9, 11, 16
7	Downloading of proteins from Protein Data Bank (PDB) and introduction to molecular modeling.	11, 16

Recommended Study Material:

1. Wilson and Giswold's Textbook of Organic, Medicinal and Pharmaceutical Chemistry,J. N. Delagado and W. A. R. Remers, Eds, J. Lipponcott Co. Philadephia.
2. Principles of Medicinal Chemistry by W. C. Foye,, Lea & Febiger, Philadelphia.
3. Burger's Medicinal Chemistry, H. E. Wolff, Ed. John Wiley & Sons, New York Oxford University Press, Oxford.
4. Smith & William's Introduction to the Principle of Drug Design and Action,4th Edition, H. John Smith, Eds,CRS Press-Taylor & Francis Group, USA.
5. Text book of Drug Design & Discovery, 3rd Edition, Povl Krogsgaard-Larsan, Tommy Liljefors & ULF Madsen, Eds, Taylor & Francis Group, USA.

6. Walter Sneader's Drug Discovery-A History, John Willy & Sons, Ltd. UK.
7. Comprehensive Medicinal Chemistry II: Volume 4: COMPUTER-ASSISTED DRUG DESIGN (Hardcover), By: Jonathan S Mason (Editor), Pub. By:Elsevier Science & Technology,2006
8. Medicinal Chemistry: An Introduction, 2nd edition, Gareth Thomas, John Wiley & Sons Ltd, England.
9. An Introduction to Medicinal Chemistry, 4th edition, Greham Patrick, Oxford University Press, NewYork.
10. The Organic Chemistry of Drug Design and Drug Action, 2nd edition, Richard B Silverman, Elsevier Academic Press.
11. QSAR and Molecular Modeling, S. P. Gupta, Anamaya Publishers, New Delhi.
12. Techniques and Experiments for Organic Chemistry, 6th edition, Adision Ault, University Science Books.
13. Vogel's Text book of Practical Organic Chemistry, ELBS/ Longman, London.
14. Practical Organic Chemistry by Mann & Saunder,, Orient Longman, London.
15. Practical Organic Chemistry by Hitesh Raval, Sunil Baldania and Dimal Shah, First Edition, Nirav & Rupal Prakashan.
16. Relevant journal articles.

DOSAGE FORM DESIGN (PH436)

(Theory & Practical)

Credits: 3 (Theory) : 2 (Practical)	Contact Hours per week: 3 (Theory) :3 (Practical)

Course Outcome:

The student will be able:

- To apply knowledge of preformulation, excipients and stability studies for designing the various dosage forms.
- To integrate the aspects of preformulation studies, stability requirements and dissolution study for the formulation of pharmaceutical products.
- To apply the knowledge of design of experiment for the development of pharmaceutical products.

Outline of course:

No.	Unit	Minimum No. of Contact Hours Approx.	Weightage %
1	Preformulation studies	10	22
2	Stability of pharmaceuticals	12	27
3	Pharmaceutical necessities (Adjuvants/Excipients)	10	22
4	Dissolution Studies	09	20
5	Design of Experiment	04	09
	Total	45	100%

Detailed Syllabus (Theory)

Sr no	Unit Details	Reference Books	No. of Hours
1	**Preformulation studies:** A. Study of physical properties of drug like physical form, particle size, shape, density, wetting, dielectric constant, solubility, dissolution & organoleptics property & their effect on formulation, stability & bioavailability. B. Crystal Characteristics: Crystalline structures, polymorphic nature of drugs, methods to identify polymorph (XRD, DSC studies) C. Drug-Excipient compatibility studies & the role of DSC and FTIR techniques to evaluate compatibility	1,4,5,13	10
2	**Stability of pharmaceuticals:** A. Kinetic principles & stability testing: Reaction rate and order. Product stability: Requirements, Shelf life, overages, containers, closures.	1,2,3,6,7	12

	B. Pathway of drug degradation and its stabilization techniques: Hydrolysis, oxidation, reduction, racemization, polymerization, acid base catalysis etc and their influence on formulation and stability of products. influence of light and temperature on stability of product. C. A Brief introduction to ICH guidelines. Accelerated stability testing procedures. MKT, climatic zones, matrixing and bracketing in stability study, real time stability. Stability protocols for various pharmaceutical products.		
3	**Pharmaceutical necessities (Adjuvants/Excipients)** Effect of following adjuvants on formulation of different pharmaceutical products: Anti- oxidants, preservatives, colours, sweetners, flavours, Diluents, binders, disintergrants, antifrictional agents, emulsifiers, suspending agents, ointment bases, solvents etc.	1, 8,10,14	10
4	**Dissolution Studies:** Introduction, theory of dissolution, apparatus, interpretation of results, Effect of micromeritics on dissolution of oral solid dosage forms, Bio relevant media. Comparison of dissolution profile by model independent (similarity and dissimilarity factor) and dependent methods like zero order, Higuchi etc.An overview of different pharmacopoeial dissolution test apparatus with respect to IP, BP & USP	9,11, 12,13	09
5	**Design of Experiments:** concept of experimental design. A brief introduction to various experimental designs and its applications	15	04

Detailed syllabus (Practical):

Sr no.	Aim of Practical
1.	Design Preformulation protocol for given drug including drug excipient compatibility studies.
2.	To evaluate different glidants for the given formula of tablet.
3.	To evaluate binders for given formula of tablet.
4.	Study the effect of diluents on tablet formulation.
5.	To evaluate the effect of different concentration of disintegrating agents on various tablet formulation
6.	To prepare calamine lotion with different concentration of suspending agents and evaluate various parameters with respect to Physical stability.

7.	Study about co-solvency for various solutes using PG, PEG & glycerin as co-solvent with water as solvent.
8.	To prepare solid dispersion of Paracetamol using PEG 4000 as carrier and study dissolution of paracetamol in pure form and from solid dispersion.
9.	Study dissolution behavior of Paracetamol tablet (marketed formulation)
10.	Study stability of ascorbic acid in given solution at given temperature.
11.	To study the Stability of aspirin /ascorbic acid at given temperature.
12.	To study the effect of pH on solubility of drugs.
13.	New Practical/s to be added by the course faculty

Recommended study material

1. The Theory & Practice of industrial pharmacy, Leon Lachman, Lea & Febiger, Varghese Publication House Bombay.
2. Remington: The Science and Practice of Pharmacy, Gennaro Alfonso R., Vol I & II, Lippincott Williams & Wilkins, New York
3. Modern Pharmaceutics, Edited by G. S. Banker & C.T. Rhodes, Marcel Dekker inc. N.Y.
4. Pharmaceutics: The science of dosage form design, Edited by M.E. Aulton. Churchill Livingstone, New York.
5. Ansel's Introduction to Pharmaceutical Dosage Form and Drug Delivry system, H.C. Ansel, N.G. Popovich, Lippincott Williams and wilkins, Philadelphia.
6. Drug Stability, edited by J. T. Cartensen, C.T.Rhode, Marcel Dekker Inc. N.Y.
7. Martin's Physical Pharmacy and Pharmaceutical Science, P.J. Sinko, Lippincott Williams & wilkins, Philadelphia.
8. Handbook of Pharmaceutical Excipients, edited by R.C. Rowe, P.J. Sheskey, M. E. Quinn, American Pharmacists Association & The Royal Pharmaceutical Society of Great Britain, USA
9. Pharmaceutical Dissolution testing, U. V. Banakar, Marcel Dekker Inc.
10. Pharmaceutical dosage forms: Tablets, H.A. Lieberman, Leon Lachman, Vol.: 1, 2, 3, Marcel Dekker inc. N.Y.
11. Indian Pharmacopoeia, published by Indian Pharmacopoeial commission, Ghaziabad
12. United State Pharmacopoeia, Indian edition, United State Pharmacopoeial convention INC.
13. British Pharmacopoeia , British Pharmacopoeia commission office, U.K.
14. Pharmaceutical Dosage Forms: Disperse system, Vol. I, II &III, **Lierberman H. A. and Leon Lachman**, Marcel Dekker, New York
15. Pharmaceutical Experimental Design , G.A.Lewis, D. Mathieu, Marcel Dekker inc. N.Y.

NOVEL DRUG DELIVERY SYSTEM (PH425)

(Theory)

Credits: 2 Contact Hours per week: 2

Course Outcome:

The student will be able:

- To have a good understanding of fundamentals - methods of preparation, evaluation of controlled drug delivery sy stems.
- To handle the novel drug delivery in the related research field.
- To describe targeted drug delivery sy stems and its significance.

Outline of the Course:

No.	Unit	Minimum No. of Contact Hours	Approx. Weightage %
1	Controlled drug delivery system	6	20
2	Oral controlled drug delivery system	6	20
3	Vesicular & Particulate Delivery systems	5	17
4	Transdermal drug delivery systems	5	17
5	Immediate release delivery systems: An overview	4	13
6	Drug targeting	4	13
Total		30	100 %

Detailed Syllabus:

No.	Unit Details	Contact hours	References
1	**Controlled Drug Delivery System:** Fundamentals of modified/controlled drug delivery systems: Fundamentals, Different Terminologies, Rational of modified release drug delivery, Advantages & Limitations, classification, factors influencing the design of controlled release dosage form including physicochemical factors.	6	4,6,7,8
2	**Oral controlled drug delivery system:** Classification, Design and development of oral controlled release dosage forms: Disolution controlled system, diffusional controlled system, , ion exchange, hydrogels, Formulation and Evaluation aspects of osmotic pressure controlled, gastro retentive delivery system	6	3,6,7,8,10
3.	**Vesicular & Particulate Delivery systems:** An Overview of Liposomes, niosomes, microparticles & nanoparticles	5	5,9

4	**Transdermal drug delivery systems:** Anatomy of skin, Basic Component of TDDS, Various approaches, Formulation using various technologies, Evaluation of TDDS, Iontophorosis, Sonophorosis, Micro needle array	5	2,3,6,7
5	**Immediate release delivery systems:** An overview Sublingual and Buccal, Oro Dispersive Tablet	4	2,10
6	**Drug targeting:** Need for drug targeting, Active & Passive targeting, various approaches for targeted drug delivery including ADEPT, Monoclonal Antibodies, Redox Mechanism, etc.)	4	5, 6,9

Recommended study materials:

1. Modern Pharmaceutics, Edited by G. S. Banker & C.T. Rhodes, Marcel Dekker inc. N.Y.
2. Pharmaceutics: The science of dosage form design, Edited by M.E. Aulton. Churchill Livingstone, New York.
3. Ansel's Introduction to Pharmaceutical Dosage Form and Drug Delivery system, H.C.
4. Ansel, N.G. Popovich, Lippincott Williams and wilkins, Philadelphia.
5. The Theory & Practice of industrial pharmacy, Leon Lachman, Lea & Febiger, Varghese Publication House Bombay.
6. Progress in Controlled & Novel Drug Delivery System, N.K.Jain , CBS Publication, New Delhi
7. Novel drug delivery systems Fundamentals & Developmental concepts by Y.W.Chien, Marcel Dekker Inc.
8. Controlled drug Delivery, Fundamentals & application by J.R. Robinson & Univent Lee, Marcel Dekker Inc.
9. Biopharmaceutics & Pharmacokinetics, D. M. Brahmankar & S. B. Jayswal, Vallabh Prakashan, New Delhi
10. Targeted & controlled drug delivery, S.P.Vyas, R.K. Khar, CBS Publisher, India
11. Encyclopedia of Pharmaceutical Technology, edited by James Swarbrik, James Braylan, Vol-1, 2,3, Marcel Dekker inc. N.Y.

Note: It is preferable to refer latest addition of all above mentioned books

Pharmacy Practice - III (PH437.1)

Credits: 3 (Practical) Contact Hours per week: 3(Practical)
Credits: 3 (Project) Contact Hours per week: 3 (Project)

Course Outcome:

The student will be able:

- To understand the different functions and services provided by the hospital pharmacy.
- To use different sources of drug information and initiate into the concept of Pharmacy practice.
- To gain knowledge and practical tools for future professional career.

List Practical:

Sr.No	Aim of Practical
1	To Familiarize with the functions of the hospital and the pharmacy department.
2	To understand different drug information resources and exercise based on it
3	To study Medication Use Processes with the help of case study
4	To study medication error and visit to hospital and evaluate cases related to different types of error
5	To study Drug Use Management and to study role of Pharmacist in formulary & DTC preparation.
6	To study drug therapy monitoring & to understand concept of ward round
7	To study concept of appropriate & rational use of medicine
8	To study main economic indicators of the hospital pharmacy
9	To study Drug information chart review and medical counseling
10	To study medical literature preparation and review for rationality

Project Guidelines:

Individual /Group projects will be given to the students. The project can be a minor research / field work type, which could be conducted in-house or the students can be deputed for the same in any recognized academic, research or clinical organization.

Sr. No	Work	Hours allotted
1	Discussion with Mentor	10
2	Experimental project /Field work	25
3	Write up, analysis of data and compilation.	10

Proposed Project Area

1. Survey based: Survey of Incidence, prevalence and occurrence of diseases or any condition (field work)
2. Meta analysis
3. Drug utilization
4. Drug safety, Pharmacovigilance
5. Pharmacoepidemiology study
6. Risk factor identification
7. Pharmacist role in Hospital (ward round)
8. Drug information bases
9. Patient Counseling
10. E Prinscribing
11. Essential drug list preparation
12. Formulary preparation, audit and comparison
13. Softwares used in Health services

14. Lab data interpretation
15. Medication error
16. Drug interaction

HERBAL FORMULATIONS-III (PH437.2)

Credits: 3 (Practical) **Contact Hours per week: 3(Practical)**
Credits: 3 (Project) **Contact Hours per week: 3 (Project)**

Course Outcome:

The student will be able:

- To prepare and evaluate herbal formulation.
- To prepare socially relevant project/seminar proposal after reviewing the literature by considering need of studies, feasibility analysis, step wise dissection of work, allocation of appropriate time span and outcome of studies.
- To work as per the plan and come out with scientific justification to the results obtained.
- To prepare a report of work done and present the project work through ICT based presentation.

Detailed Syllabus Project: (03 Hr. /Week)

Project Guidelines:
Individual /Group projects will be given to the students. The project can be a minor research / field work type, which could be conducted in-house or the students can be deputed for the same in any recognized academic, research or clinical organization.

Sr. No	Work	Hours allotted
1	Discussion with Mentor	10
2	Experimental project /Field work	25
3	Write up, analysis of data and compilation.	10

The project will be based on following topics;

1. Preparation and Evaluation of Herbal Formulation and Herbal Cosmetics
2. Comparative Evaluation of Marketed Herbal Formulations
3. Market Survey on Herbal Formulation and Herbal Cosmetics
4. Branding and Marketing of Herbal Formulations
5. Clinical and Preclinical Reviews of herbal Drugs and Herbal Formulations
6. Regulatory status of Herbal Formulations in Various Countries

Practicals to be offered:

No.	Aim of the Practical
1	Formulation development & standardization of herbal cream.
2	Formulation development & standardization of herbal gel.
3	Formulation development & standardization of herbal shampoo.
4	Formulation development & standardization of herbal toothpaste.
5	Formulation development & standardization of herbal tablet.
6	Formulation development & standardization of herbal syrup.
7	Evaluation of marketed Ayurveda formulation.

Recommended study materials:

1. Herbal Drug Industry. Chief Editor R.D. Chaudhary, 1st Edition, 1996. Eastern Publishers, New Delhi.

2. The ayurvedic Formulary of India, Part I and II, Department of Indian system of Medicine and Homoeopathy, Govt, of India, 1st English Edition, 2000.
3. Herbal drug technology, S. S. Agrawal and M. Paridhavi, Univeristies Press, 1st Edition, 2007.
4. Handbook of Cosmetic Science and Technology, Andre Barel, Marc Paye, Howard I. Maibach, CRC Press.
5. Cosmetic technology, Nanda S, Nanda A, Khar RK., Birla Publications Pvt. Ltd
6. Cosmetics: Science and Technology, Balsam S.M. and Sagarin Edward, 2nd Ed, Wiley Interscience.

Production of Bulk Drugs and Drug Intermediates III (PH437.3)

Credits: 3 (Practical) **Contact Hours per week: 3 (Practical)**
: 3 (Projects) **: 3 (Projects)**

Course Outcome:

The student will be able:

- To explain basic concept of Process Chemistry along with Unit process for bulk drug manufacturing.
- To demonstrate various techniques and steps involve in a bulk drug manufacturing in pharmaceutical industries.
- To describe the significance of WHO good manufacturing practices for active pharmaceutical ingredients.
- To describe Quality by Design (QbD) approch for manufacturing of Active Pharmaceutical Ingredients.

Detailed Syllabus of Project: (03 Hr. /Week)

Project Guidelines:

Individual /Group projects will be given to the students. The project can be a minor research / field work type, which could be conducted in-house or the students can be deputed for the same in any recognized academic, research or clinical organization.

Sr. No	Work	Hours allotted
1	Discussion with Mentor	10
2	Experimental project /Field work	25
3	Write up, analysis of data and compilation.	10

The project will be based on following topics;

1. Solvent Purification Techniques
2. Crystallization Techniques
3. Structural Elucidation of Organic Compounds
4. Bulk Drug Process and Manufacturing: Case Study
5. Application of Green Chemistry
6. Application of Microwave Synthesis and its advantages over Conventional synthesis
7. Importance of QbD and DOE in API Manufacturing
8. Stereochemistry and Chiral Separation of Enantiomer
9. Synthon Approach
10. Protection and Deprotection Approach
11. Synthesis of API's (Laboratory Work)

Detailed Syllabus (Practical):(03 Hr./Week)

Topic No.	**Aim of the Practical**
1	Synthesis of Aspirin form salicylic Acid by Common synthetic route and Microwave Approach and compare the Practical yields.
2	Synthesis by Common synthetic route and reaction in aqueous medium and compare the Practical yields.
3	Use of different solvents for Synthesis (01 Practical)
4	Case study base practical (Bulk Drug Production Process of any two API's)
5	Reaction in Aqueous Medium (01 Practical)

6	Solvent Recovery Techniques (01 Practical)
7	Reaction by use of different Catalyst (Esterification) (01 Practical)
8	Crystallization from different solvents (02 Practical)
9	Structural Elucidation of Organic Compounds (02 Practical's)

References:

1. Vogel's Text Book of Practical Organic Chemistry- Brian Furniss, Antony Hannaford,
   Peter Smith, Austrin (Eds), 5th edition, ELBS Publication, Singapore, 1997.
2. Process Chemistry in the Pharmaceutical Industry, Volume 1edited by Kumar Gadamasetti
3. N. G. Anderson, ''Practical Process Research & Development'', Academic Press 2000.
4. O. Repic, ''Principles of Process Research and Chemical Development in the Pharmaceutical Industry, Wiley 1998.
5. N. Yasuda, "The Art of Process Chemistry", Wiley 2010
6. R.A. Sheldon, I. Arends, U. Hanefeld, ''Green Chemistry & Catalysis'', Wiley 2007

Pharmaceutical Quality Assurance and Quality Control – III (PH437.4)

Credits : 3 (Practical) Contact Hours per week : 3 (Practical)
: 3 (Project) : 3 (Project)

Course Outcome:

The student will be able:

- To prepare SOP, Validation protocols and validation reports.
- To perform in-process quality control tests for tablets and capsules.
- To perform calibration of analytical instruments.

Detailed Syllabus (Practical):

Sr. No	Aim of the Practical
1	To perform IPQC tests for Tablets
2	To perform IPQC tests for capsules
3	To calibrate pH meter, analytical balance, volumetric glassware.
4	To calibrate UV spectrophotometer
5	To perform validation of autoclave and oven
6	To prepare SOP of any two analytical equipments
7	To prepare SOP of any two processing equipments
8	To prepare validation protocol.
9	To prepare validation report
10	To prepare record sheet
11	Case studies on Quality Risk Management, GMP non-compliance, Regulatory Audit
12	Any other relevant to topics covered in theory course 1 and 2 of elective on pharmaceutical quality assurance and quality control

Project:

Project Guidelines:

Individual /Group projects will be given to the students. The project can be a minor research / field work type, which could be conducted in-house or the students can be deputed for the same in any recognized academic, research or clinical organization.

Sr. No	Work	Hours allotted
1	Discussion with Mentor	10
2	Experimental project /Field work	25
3	Write up, analysis of data and compilation.	10

Students are expected to pursue a project in the broad area of quality assurance and quality control. It may be specifically on development of protocol, preparation of standard operating procedure, validation of equipment, validation of process, validation of facilities, development and validation of analytical procedure, calibration master plan, Manufacturing Risk management, pharmaceutical quality system or any other area relevant to pharmaceutical quality assurance and quality control. The project may be individual or in group.

Recommended study materials:

1. Total Quality Management- Guiding Principle for Application, J. P. Peker, ASTM manual series, Philadelphia.
2. Total Quality Management – The Key to Business Improvemetn, Champman & Hall, Londan.
3. Quality Assurance Guide by Organisation of Pharmaceutical products of India.
4. A guide to Total Quality Management – Kaushik Maitra and Sedhan K.Ghosh.

5. ISO 9000 and Total Quality Management – Sadhank. G. Ghosh.
6. Quality Assurance of Pharmaceuticals – A compendium of guidelines and related materials Vol.1and Vol.2, WHO, (1999).
7. ICH guidelines

DRUG REGULATORY AFFAIRS- III (Elective-III) (PH437.5)

Credits: 3(Practical) **Contact Hours per week: 3(Practical)**

Credits: 3 (Project) **Contact Hours per week: 3(Project)**

Course Outcome:

The student will be able:

- To have knowledge of all the regulatory agencies of the world.
- To prepare the documents like IPQC protocol, BMR, MFR, CTD, Specifications, etc.
- To prepare the documents of Pharmaceutical Regulatory.

List of Practicals

- To prepare Master formula record (MFR) and study Batch formula record (BFR) for any two formulations.
- To prepare in process quality control (IPQC) protocol for any two formulations.
- To study and understand components of clinical research protocols.
- To prepare sample stability study protocol for any two formulations.
- To prepare sample Preclinical study reports and summaries.
- To prepare package design and package inserts for any two formulations.
- To study Specification and Method of analysis protocol for any one API.
- To prepare Analytical method validation protocol.
- To prepare modules of CTD document as per guidelines of regulatory agencies of different countries.
- To analyze cases related to the regulatory affairs.
- To study sample dossier for getting approval of solid dosage form (national and/or International) and to write dossier in group of 4-5 on any selected drug product.

Project Guidelines:

- Individual /Group projects will be given to the students. The project can be a minor research / field work type, which could be conducted in-house or the students can be deputed for the same in any recognized academic, research or clinical organization.

Sr. No	Work	Hours allotted
1	Discussion with Mentor	10
2	Experimental project /Field work	25
3	Write up, analysis of data and compilation.	10

List of Projects

- Preparation of various document/s for registration of drug product in India.
- Preparation of drug master file for registration of drug in India, USA & EU.
- Analysis of cases related to the regulatory affairs.
- Preparation of modules as per ASEAN, ICH & WHO guidelines.
- Preparation of dossiers as per guidelines of regulatory agencies of different countries.

References:

1. Medical Product Regulatory Affairs: Pharmaceuticals, Diagnostics, Medical Devices By John J. Tobin and Gary Walsh
2. FDA Regulatory Affairs: A Guide for Prescription Drugs, Medical Devices, and Biologics, Second Edition by Douglas J. Pisano and David S. Mantus

3. Good Drug Regulatory Practices: A Regulatory Affairs Quality Manual (Good Drug Development Series, Vol. 1 by Helene I. Dumitriu
4. CGMP-a good manufacturing practice for pharmaceuticals, Manohar Potdar, Pharmamed press, 2008.
5. Validation in Pharmaceutical Industry (Concept, Approaches & Guidelines), P.P.Sharma, Vandana Publications, New Delhi.
6. Official website of regulatory agencies of different countries.

STERILE PRODUCTS TECHNOLOGY-III (PH437.6)

Credits: 3(Practical) Contact Hours per week: 3(Practical)

Credits: 3 (Project) Contact Hours per week: 3(Project)

Course Outcome:

The student will be able:

- To learn formulation and development aspects of various sterile products.
- To understand the concept of Lyophilization and critical parameters to be taken into consideration while developing a freeze drying cycle.
- To impart the technical skills for manufacturing of Sterile Preparations on Lab Scale.

List of Practical:

- To prepare and evaluate 5% dextrose injection.
- To prepare and evaluate 0.5% ciprofloxacin eye drop.
- To demonstrate various aspects of environmental monitoring.
- To demonstrate freeze drying process and freeze drying cycle development.
- To develop freeze drying cycle for 5% Dextrose Solution.
- Freeze drying of Pharmaceutical product: To study the Annealing Phenomena.
- To study the Effect of freezing rate on Pharmaceutical product during Lyophilization process.
- To study the effect of different Lyo-protectant on Pharmaceutical product during Lyophilization process.
- To study the effect of different various concentration of Lyo-protectant on Pharmaceutical product during Lyophilization process.
- To evaluate marketed sterile implant/device product.
- To Design and develop of peptides, proteins & Medical device

Project Guidelines:

Individual /Group projects will be given to the students. The project can be a minor research / field work type, which could be conducted in-house or the students can be deputed for the same in any recognized academic, research or clinical organization.

Sr. No	Work	Hours allotted
1	Discussion with Mentor	10
2	Experimental project /Field work	25
3	Write up, analysis of data and compilation.	10

List of Projects:

- Study of different Lyo-protectant on Pharmaceutical product during Lyophilization process.
- Study of marketed sterile implant/device product.
- Design and development of peptides and proteins
- Design and development of Medical device
- Formulation, evaluation and packaging of parenteral products
- Formulation, evaluation and packaging of ophthalmic products

Recommended Study Materials:

1. The Theory & Practice of industrial pharmacy, Leon Lachman, Lea & Febiger, Varghese Publication House Bombay.
2. Remington's Pharmaceutical Sciences , Edited by A. R. Gennaro, Mack Publishing Co,
3. Pharmaceutical dosage forms: Parenteral, Vol.: 1, 2, 3, Leon, Lachman, K. E. Avis, Marcel Dekker inc. N.Y.
4. Indian Pharmacopoeia , published by Indian Pharmacopoeial commission, Ghaziabad

5. United State Pharmacopoeia, United State Pharmacopoeial convention INC.
6. British Pharmacopoeia , British Pharmacopoeia commission office, U.K.
7. Handbook of Pharmaceutical Manufacturing Formulations: Sterile Products, Sarfaraz K. Niazi Volume 6, Informa Healthcare USA, Inc.2009.
8. Encyclopedia of Pharmaceutical Technology, Jasmes Swarbrick and James C. Boylan, Marcel Dekker Inc., New York.
9. Achieving Sterility in Medical and Pharmaceutical Products, Nigel A. Halls, volume 64.
10. Development of Biopharmaceutical Parenteral Dosage Forms, edited by John A. Bontempo, volume 85.
11. Parenteral Quality Control: Sterility, Pyrogen, Particulate, and Package Integrity Testing: Third Edition, Revised and Expanded, Michael J. Akers, Daniel S. Larrimore, and Dana Morton Guazz, volume 125.
12. Ophthalmic Drug Delivery Systems: Second Edition, Revised and Expanded, edited by Ashim K. Mitra, volume 130.
13. Injectable Dispersed Systems: Formulation, Processing, and Performance, edited by Diane J. Burgess, volume 149.
14. Environmental Monitoring for Cleanrooms and Controlled Environments, edited by Anne Marie Dixon, volume 164.
15. Freeze-Drying/Lyophilization of Pharmaceutical and Biological Products: Second Edition, Revised and Expanded, edited by Louis Rey and Joan C. May, volume 137.

ENTREPRENEURSHIP FOR PHARMA PROFESSIONALS (PH438)

(Theory & Practical)

Credit: 3 (Theory) : 1 (Practical)	Contact hours per Week: 3 (Theory) : 3 (Practical)

Course Outcome:

The student will be able:

- To understand Entrepreneurship as a desirable and feasible career option.
- To build the necessary competencies and motivation for a career in entrepreneurship.

Outline of the Course:

No.	Unit	Minimum no. of contact hour	Appprox. Weightage
1	Entrepreneur-Entrepreneurship – Enterprise - Intrapreneurship	9	20
2	Opportunity scouting and idea generation	9	20
3	Entrepreneurial Challenges in small business	9	20
4	Governments Role in Growth of MSMEs & SMEs	6	13
5	Enterprise Scale Up, Growth & Harvesting	6	13
6	Case studies on successful and unsuccessful entrepreneurs and ventures	6	14
	Total	45	100

Detailed syllabus (Theory):

Sr. No.	Unit Details	No. of hours
1	**Entrepreneur-Entrepreneurship-Enterprise- Intrapreneurship:** • Conceptual issues. • Entrepreneurship vs. Management. • Role of entrepreneurs in relation to the enterprise and in relation to the economy. • Entrepreneurship as an interactive process between the individual and the environment. • Small business as the seedbed of entrepreneurship. • Entrepreneurial competencies. • Entrepreneurial motivation, performance and rewards. • Scope and Need of NGO's in Pharmaceuticals • Growth drivers in Indian Pharmaceutical Industry • Import and Export in Pharma Industry • Business ethics related to Pharmaceutical industry	9
2	**Opportunity scouting and idea generation:**	9

	• Role of creativity & innovation and business research. • Sources of business ideas. • Concept of Technopreneur – Technology based Enterprise (products protected by patent taken up for commercialization) • Entrepreneurial opportunities in contemporary business environment, for example opportunities in pharmaceuticals • The process of setting up a small business. (Legitimate compliances) • Business Plan- contents, importance, pre-requisites for an effective business plan. • Basic of market research, Introduction, different method of research, target group, etc. • Company Start up either Pharma marketing or Manufacturing	
3	**Entrepreneurial Challenges in small business** • Sources of venture capital, fixed capital, working capital and a basic awareness of financial services such as venture capital, private equity, leasing and factoring. • The concept and application of product life cycle, advertising & publicity, sales & distribution management. • HR challenges- Availability of talent, retention, compensation, etc. • Legal and Government Challenges. • Licensing Process in Pharma Manufacturing and Marketing	9
4	**Governments Role in Growth of MSMEs & SMEs** Schemes from Central Governments, State Governments and Facilitating institutions in the sectors of • Chemicals • Healthcare • Pharmaceuticals Govt. schemes to encourage Entrepreneurship. • Eg. Subsidies/Tax exemption/Financial help/Export benefits.	6
5	**Enterprise Scale Up, Growth & Harvesting** A) Methods and Ways to take the small enterprise to large scale Organizations. B) Funding Options: • Bank Loans/Other Private Funding • Listing on stock exchanges • Partnership & Alliances – Global Companies • Other Alternatives C) Harvesting the correct potential of Enterprise.	6
6	**Case studies on successful and unsuccessful entrepreneurs and ventures**	6

Recommended Books/Journals/Website:

1. Brandt, Steven C., The 10 Commandments for Building a Growth Company, Third Edition, Macmillan Business Books, Delhi, 1977.
2. Dollinger, Mare J., Entrepreneurship: Strategies and Resources, Illinois, Irwin, 1955.
3. Entrepreneurial Business Law Journal

4. Entrepreneurship Theory and Practice
5. Holt, David H., Entrepreneurship: New Venture Creation, Prentice-Hall of India, latest Edition.
6. http://msmestartupkit.com
7. http://www.dcmsme.gov.in
8. Journal of Business Venturing
9. Journal of Developmental Entrepreneurship
10. Journal of International Entrepreneurship
11. Journal of Small Business Management
12. Journal: Entrepreneurship and Regional Development.
13. Kazmi, Azhar, “What Young Entrepreneurs Think and Do: A Study of Second Generation Business Entrepreneurs,” The Journal of Entrepreneurship, 8, No. 1, 1999, pp. 67-78.
14. Small Business Economics
15. Vesper, KarlsH, New Venture Strategies, (Revised Edition), New Jersey, Prentice-Hall, 1990.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF PHARMACY
RAMANBHAI PATEL COLLEGE OF PHARMACY
B. PHARM PROGRAM
COURSE ARTICULATION MATRIX

Course Code	Course Name	Course Outcome	PO1 (Pharmacy Knowledge)	PO2 (Planning Abilities)	PO3 (Problem Analysis)	PO4 (Modern Usage tools)	PO5 (Leadership Skills)	PO6 (Professional Identity)	PO7 (Pharmaceutical Ethics)	PO8 (Communication)	PO9 (The Pharmacist and Society)	PO10 (Environment and Sustainability)	PO11 (Life-Long Learning)
Semester-7													
PH430	Clinical Pharmacy and Pharmacotherapeutics-I												
	CO1	Describe the concept of Clinical Pharmacy & Pharmacotherapy	3	2	-	-	-	2	-	-	-	-	
	CO2	Able to describe role of clinical pharmacist as a part of health care team.	3	2	-	-	2	-	-	-	-	-	1
	CO3	Able to describe the theoretical considerations and basic principles of pharmacotherapeutics in the management of the diseases and conditions of various body system such as CVS, renal , hematology and malignant system with the help of standard treatment guidelines and treatment algorithms.	3	2	2	-	-	-	-	-	-	-	1
	CO4	Able to Recognize the diagnostic criteria required to specific disease as well as any clinical testing requirements for monitoring drug effectiveness and potential toxicity.	3	2	2	2	-	-	-	-	-	-	1
PH431	Traditional Medicine and Herbal Drugs Technology												
	CO1	Student should be able to understand basic concept of Ayurveda and describe composition and methods of preparation of different ayurvedic formulations.	3		1	-	-	-	-	-	3	-	3
	CO2	Student should be able to describe and compare pharmacognostic study of Indian traditional drugs.	3		3	-	-	-	-	-	2	-	1
	CO3	Student should be able to understand and describe various concepts and components of plant tissue culture.	3		3	-	-	-	-	-	-	-	3
	CO4	Student should be able to apply knowledge in quality control, standardization of crude drugs as per WHO guideline	3		3	-	-	-	-	-	-	-	3

	CO5	Students should be able to describe various natural allergens and photosensitizing agents.	3		3	-	-	-	-	-	-	-	3
PH432	Pharmaceutical Analysis-III												
	CO1	Apply appropriate spectroscopic and thermal methods for analysis of pharmaceuticals based on basic understandings of technique.	3	2	3	3	-	-	-	-	-	-	3
	CO2	Describe instrumentation and operating principle of spectroscopic and thermal techniques	3			3	-	-	-	-	-	-	3
	CO3	Interpret the spectroscopic data and establish the identity of the compound	3	1	3	3	-	-	-	-	-	-	3
	CO4	Understand basic concepts and components of Quality Control, Quality Assurance, Good Manufacturing Practices and Good Laboratory Practices.	3			2	-	-	-	-	-	-	3
	CO5	Independently process, interpret the data obtained through experimentation and report the results.	3	1	3	3	-	-	-	1	-	-	3
PH418	Medicinal Chemistry-III												
	CO1	Students would be able to understand the general synthetic pathways of drugs (Theoretically/Practically) it's therapeutics use, including its chemical classification, therapeutics uses and mechanism of action under the classes of chemotherapeutic agents and antibiotics.	3	-	-	-	-	-	-	-	-	-	-
	CO2	Able to understand structural activity relationship of medicinal class of drugs.	3			1	-	-	-	-	-	-	1
	CO3	Students would be able to understand a novel appproach like" Microweave and Green Chemistry for synthesis of various chemical entities.	3		2	1	-	-	-	-	-	2	1
PH433	Pharmaceutical Technology-II												
	CO1	Students would be able to employ the knowledge of pharmaceutical technology used in production of sterile dosage forms, liquid dosage forms, aerosol and cosmetic products with good manufacturing practices.	3	2	2	1		1	1	2	-	-	1
	CO2	Students would be able to describe in detail instrumentation/equipments for formulation and evaluation of pharmaceutical products as per standards.	3	2	2	1	1	1	1		-	-	1
	CO3	Students would be able to explain the fundamentals of pilot plant scale up techniques.	3	1	1	1	2		1	2	-	-	1
PH420.1	Pharmacy Practice - II												
	CO1	Students would be able to learn the concept of health economics.	3	1	1	-	-	-	-	-	-	-	1
	CO2	Students would be understand the role of the pharmacist in Ethics and roles in Ethical issues in pharmacy practice research.	3	-	-	-	-	-	-	-	-	-	1

	CO3	Students would be able to develop skills of research in clinical pharmacy practice.	3	1	1	-	-	1	1	-	-	-	1
PH420.2	Herbal Formulations- II												
	CO1	Students should be able to summarize different types preformulation studies and their significance in formulation designing	3	1	2	1	-	-	-	-	-	1	3
	CO2	Students will be able to develop formula of the selected formulations containing herbal extracts with scientific justifications.	3	2	2	1	-	-	-	-	-	1	3
	CO3	The students will be able to decide the quality attributes to be assessed for selected formulations	3	1	2	1	-	-	-	-	-	1	3
PH420.3	Production Of Bulk Drugs And Drug Intermediates II												
	CO1	Explain basic concept of Process Chemistry along with Unit process for buld drug manufacturing.	3		2	1	-	-	-	-	-	1	3
	CO2	Demonstrate various techniques and steps involve in a buld drug manufacturing in pharmaceutical industries.	3	2	2	1	-	-	-	-	-	1	3
	CO3	Students would be able to understand various environmentally safe synthetic pathway for synthesis of drug intermediates.	3	1	2	1	-	-	-	-	-	1	3
	CO4	Describe the process for product registration.											
PH420.4	Pharmaceutical Quality Assurance And Quality Control - II												
	CO1	Elaborate the role of validation in assurance of quality of pharmaceutical industries.	3	-	-	-	2	-	-	-	-	-	3
	CO2	Describe various aspects of documentation – SOPs, records and reports required for pharmaceutical Industries.	3	-	-	-	-	-	-	-	-	-	3
	CO3	Explain the types, approaches and reports of quality inspections and audits.	3	-	-	-	-	-	-	-	-	-	3
PH420.5	Drug Regulatory Affairs- II												
	CO1	Students should be able to understand the importance of WHO guidelines for the drugs and Pharmaceuticals relevant to the regulatory filing.	1	-	1		-	-	-	-	2	2	1
	CO2	Students should be able to understand the importance of guidelines of International Conference on Harmonization (ICH) relevant to the regulatory filing.	1	-	-	-	-	1	-	-	1	-	-
	CO3	Students should be able to prepare a dossier as per the guidelines of ASEAN countries.	2	-	1	-	-	1	1	1	-	2	1
PH420.6	Sterile Product Technology- II												
	CO1	Students should be able to understand the various routes of administration and the other anatomical aspects for the administration of sterile dosage forms.	3	-	-	-	-	-	-	-	-	-	3

	CO2	Students shouldbe able to understand applications of sterile dosage forms in manufacturing and packaging of drugs/dosage forms.	3	-	-	-	-	-	-	-	-	-	3
	CO3	Students should familiar with various sterile dosage forms and its applications.	3	-	3	-	-	-	-	-	-	-	3
PH421	Environmental Sciences												
	CO1	Understand the natural environment and its relationships with human activities.	-	1	-	-	-	-	-	-	-	-	-
	CO2	Explain the methods for conservation of natural resources.	-	-	-	-	-	-	-	-	-	2	1
	CO3	Integrate the facts, concepts and methods to environmental issues in pharmaceutical industry.	3	1	-	1	-	-	-	-	1	-	-
Semester-8													
PH434	Clinical Pharmacy and Pharmacotherapeutics-II												
	CO1	able to counsel the patient in terms of patient compliance and appropriate drug use	3	2	-	-	2	-	-	-	-	-	-
	CO2	Discuss the concept of rational prescribing & essential drug	3	2	-	-	-	-	2		2	-	-
	CO3	Able to describe the theoretical considerations and basic principles of pharmacotherapeutics in the management of the diseases and conditions of various body system such as CNs, mascular system, endocrine , infectious and respiratory system with the help of standard treatment guidelines and treatment algorithms.	3	2	3	2	-	-	-	-	-	-	2
	CO4	Able to discuss the importance of individualized therapeutic plans based on patients condition	3	2	2	2	-	-	-	-	-	-	2
PH435	Medicinal Chemistry-IV												
	CO1	Students would be able to understand the physicochemical properties of drugs and Structure Activity Relationship (SAR)in relation to drug design and discovery.	3	-	1	-	-	-	-	-	-	-	1
	CO2	Students would be able to describe their knowledge and understanding of computational and conventional techniques applicable to drug discovery.	3	-	1	2	-	-	-	-	-	-	1
	CO3	Students would be able to have knowledge and understanding of combinatorial chemistry techniques applicable to drug design and discovery.	3	-	-	1	-	-	-	-	-	-	-
	CO4	Students would be able to apply earned theoritical knowledge to their experimental aspects for developing of drug candidate.	3	-	-	-	-	-	-	-	-	-	-
PH436	Dosage Form Design												
	CO1	To apply knowledge of preformulation, excipients and stability studies for designing the various dosage forms.	3	1	1	1	-	1	1	1	-	-	1
	CO2	The students would be able to integrate the aspects of preformulation studies, stability requirements and dissolution study for the formulation of pharmaceutical products.	3	1	1	1	-	1	1	1	-	-	1

	CO3	The students would be able to apply the knowledge of design of experiment for the development of pharmaceutical products.	3	1	1	1	-	1	1	1	-	-	1
PH425	Novel Drug Delivery Systems												
	CO1	The course would help the student to have a good understanding of fundamentals - , methods of preparation, evaluation of controlled drug delivery sy stems	3	-	-	1	1	1	1		1	-	1
	CO2	The knowledge gained by the students during the study of this course can also help them in handling of novel drug delivery in the related research field	3	1	1	1		1	1	2	1	-	1
	CO3	Students can describe targeted drug delivery sy stems & its significance.	3		1	1		1	1		1	-	1
PH437.1	Pharmacy Practice - III												
	CO1	The course would help to understand the different functions and services provided by the hospital pharmacy	3	-	1	-	-	1	-	1	-	-	1
	CO2	The course would help students to use different sources of drug information and initiate into the concept of Pharmacy practice	3	-	1	-	-	1	-	1	-	-	1
	CO3	The course would help to gain knowledge and practical tools for future professional career.	3	1		-	-	1	-	1	-	-	1
PH437.2	Herbal Formulations- III												
	CO1	Students should be able to prepare and evaluate herbal formulation	3	-	3	-	-	-	-	-	-	-	3
	CO2	Students will be able to prepare socially relevant project/seminar proposal after reviewing the literature by considering need of studies, feasibility analysis, step wise dissection of work, allocation of appropriate time span and outcome of studies.	3	-	3	-	-	-	-	-	2	-	3
	CO3	Student will be able to work as per the plan and come out with scientific justification to the results obtained.	3	-	3	-	-	-	-	-	-	-	3
	CO4	Students will be able to prepare a report of work done and present the project work through ICT based presentation.	3	-	3	-	2	-	-	-	-	-	3
PH437.3	Production Of Bulk Drugs And Drug Intermediates III												
	CO1	Explain basic concept of Process Chemistry along with Unit process for buld drug manufacturing.	3	-	-	-	-	-	-	-	-	-	3
	CO2	Demonstrate various techniques and steps involve in a buld drug manufacturing in pharmaceutical industries.	3	-	-	3	-	-	-	-	-	-	3
	CO3	Describe the significance of WHO good manufacturing practices for active pharmaceutical ingredients.	3	-	2	-	-	-	-	-	-	-	3
	CO4	Describe Quality by Design (QbD) approch for manufacturing of Active Pharmaceutical Ingredients.	3	-	-	3	-	-	-	-	-	-	3
PH437.4	Pharmaceutical Quality Assurance And Quality Control - III												
	CO1	Prepare SOP, Validation protocols and validation reports.	3	3	1	1	-	-	-	1	-	-	3

	CO2	Perform in-process quality control tests for tablets and capsules.	3	3	1	3	-	-	-	-	-	-	3
	CO3	Perform calibration of analytical instruments.	3	3	2	3	-	-	-	-	-	-	3
PH437.5	Drug Regulatory Affairs- III												
	CO1	The student should have knowledge of all the regulatory agencies of the world.	1	-	-	-	-	1	1		1	1	-
	CO2	The should be able to prepare the documents like IPQC protocol, BMR, MFR, CTD, Specifications, etc. -	2	-	1	1	1	-	-	-	-	-	-
	CO3	student should be able to prepare the documents of Pharmaceutical Regulatory.	2	1	1	-	-	1	-	1	1		2
PH437.6	Sterile Product Technology - III												
	CO1	Students should able to learn formulation and development aspects of various sterile products	3	-	-	3	-	-	-	-	-	-	3
	CO2	Students should able to understand the concept of Lyophilisation and critical parameters to be taken into consideration while developing a freeze drying cycle.	3	-	-	3	-	-	-	-	-	-	3
	CO3	Students should impart the technical skills for manufacturing of Sterile Preparations on Lab Scale.	3	-	-	3	-	-	-	-	-	-	3
PH438	Entrepreneurship for Pharma professionals												
	CO1	Entrepreneurship as a desirable and feasible career option	-	-	3	-	-	-	-	3	2	-	1
	CO2	Build the necessary competencies and motivation for a career in entrepreneurship	-	-	3	-	-	-	-	3	2	-	1